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BOK-LOOK: A Semi-Structural Model for Korea's Open Economy and Monetary Policy Analysis

This paper introduces BOK-LOOK, a large-scale, semi-structural model developed by the Bank of Korea to support its forecasting and policy analysis. Inspired by the FRB/US model, it is estimated using a fully Bayesian method to ensure robust parameter estimation and empirical consistency. The model employs a VAR-based expectations framework and polynomial adjustment cost specifications to balance theoretical coherence with empirical adaptability. The model features a detailed external sector by the inclusion of a high-resolution foreign blocks and financial synchronization channels that captures the impact of the global business and financial cycle to Korean economy. Consequently, BOK-LOOK is tailored to generate internally consistent projections for key macro variables and supports various scenario analyses. Another distinctive feature is its comprehensive treatment of financial stability concerns through the explicit incorporation of household debt and housing prices dynamics, reflecting Korea's unique economic and financial characteristics. By providing both economic forecast and counterfactual analysis capabilities, the model establishes a coherent framework for understanding macroeconomic dynamics and informing monetary policy decisions.

Keywords: FRB/US, Semi-structural Modeling, Forecasting, Scenario Analysis, Monetary Policy

JEL Classification: C3, E17, E37, E52, E58

Ⅰ. Introduction

The global economic environment became increasingly complex and uncertain, marked by significant disruptions such as the Great Recession and the COVID-19 pandemic. These crises reshaped the dynamics of global trade, financial flows, and policy responses, presenting central banks with the critical task of adapting their analytical frameworks to new economic realities. For the Bank of Korea (BOK), additional challenges stem from domestic factors, including heightened household debt, fluctuations in housing prices, and an aging population. Furthermore, Korea's status as a small open economy makes it highly sensitive to external conditions, such as changes in U.S. monetary policy stance and global value chains.

This paper introduces BOK-LOOK, the Bank of Korea's Large Outlook model Of Korea. As a large-scale, semi-structural macroeconomic model, BOK-LOOK is inspired by the Fed's FRB/US model (Brayton and Tinsley 1996), which became a benchmark for central banks seeking to balance theoretical coherence with empirical adaptability. FRB/US-type models are characterized by their ability to represent long-run economic relationships grounded in economic theory while accommodating short-run dynamics influenced by real-world frictions. By augmenting error-correction frameworks with optimization-based, forward-looking adjustments, these models effectively capture the interplay between policy instruments and macroeconomic outcomes.

BOK-LOOK builds upon this foundation while addressing the specific needs of Korea's economy. One of its key features is a high-resolution external block that captures detailed interactions between Korea and its global trading partners. By modeling these relationships, the model provides insights into the transmission of external shocks through trade and financial channels. Another distinguishing feature is the inclusion of a financial sector block that explicitly incorporates household debt, housing prices, and monetary policy transmission. This addition enables the model to analyze critical issues related to financial stability, which is essential in Korea's economic and financial context.

The motivation for developing BOK-LOOK also aligns with recent advancements in central bank modeling practices. Bernanke (2024) highlights the need for central bank models to incorporate rich and institutionally realistic representations of monetary policy transmission, empirically accurate descriptions of pricing dynamics, and detailed modeling of key sectors such as housing and finance. BOK-LOOK adopts these principles, ensuring that it not only meets international standards but also addresses Korea's unique economic characteristics.

BOK-LOOK is composed of four main blocks—external, expenditure, price, and financial—encompassing more than 140 endogenous variables and 200 equations. These blocks are interconnected, enabling the model to capture the complex interactions.

Preliminary results demonstrate the model's ability to generate consistent and theoretically grounded forecasts for key macroeconomic variables, including GDP growth, inflations, and policy rates. We also show that the key impulse responses from the model are in the line of consensus views of BOK on the Korean macroeconomy. These capabilities highlight its potential as a versatile tool for forecasting and policy analysis.

BOK-LOOK builds on the foundational traits of the FRB/US model (Brayton and Tinsley 1996), but with modifications tailored to reflect the unique economic conditions of Korea. These adaptations enhance its ability to address the challenges of an interconnected global economy while providing practical tools for policy analysis and forecasting.

One of the most important contributions of BOK-LOOK lies in its detailed representation of the foreign sector. The model effectively captures the interactions between Korea's key trading partners while integrating these dynamics into the domestic economy. This design underscores Korea's deep economic ties with major global economies.

Additionally, it also captures financial synchronization between international and domestic markets. Unlike fully structural models such as DSGE frameworks, which often face challenges in representing the connections between real and financial sectors, BOK-LOOK's semi-structural design offers greater flexibility. The model explicitly captures how U.S. long-term interest rates influence Korea's domestic long-term rates via term premiums, highlighting strong global financial interdependence.

By considering both direct trade linkages and regional spillover effects, the model serves as a tool for analyzing external shocks. For instance, the model can assess the implications of U.S. monetary policy adjustments or economic slowdowns in China on Korea’s export and import activities. These capabilities are essential for understanding how global economic trends ripple through Korea’s open economy.

Another key feature of BOK-LOOK is that it can serve as a framework for studying financial stability. The financial block of the model incorporates household debt, and housing prices. This allows policymakers to assess the dual impacts of monetary policy on the real economy and financial stability, making it particularly valuable for managing Korea’s highly leveraged household sector.

Finally, BOK-LOOK introduces optional toolkits to support conditional forecasting and scenario analysis. By enabling a subset of endogenous variables to be fixed, the model ensures consistency with the bank’s view on economic forecasts represented by judgments or other models. These toolkits allow for scenario analyses that explore the effects of changes in external conditions, such as growth and inflation impacts resulting from variations in global financial trends or trade policies. This flexibility makes BOK-LOOK a practical and adaptable framework for navigating uncertainty in the global economic environment.

BOK-LOOK has been developed as a cornerstone of efforts to modernize Korea’s economic forecasting and advance the Bank of Korea’s monetary policy framework. The model is designed to provide stable medium-term forecasts ensuring that the policy rate projection aligns with macroeconomic fundamentals such as the business cycle and inflation dynamics. By generating forecasts where key variables are endogenously and coherently determined within the model’s structure, BOK-LOOK offers a reliable baseline for monetary policy planning.

Additionally, BOK-LOOK facilitates in-depth scenario and policy effect analyses, enabling policymakers to explore the implications of various economic shocks and policy adjustments. By offering both stable forecasts and flexible scenario analyses, the model supports the Bank of Korea’s goal of enhancing its forecasting system against the dynamic economic environment.

Ⅱ. Literature Review

Our model is definitely positioned in the stream of the FRB/US type model originated by Brayton and Tinsley (1996), a modern succession of traditional simultaneous equation systems.[1] These semi-structural models seek both goodness-of-fit and theoretical interpretability. What sets the model apart to its ancestors comes from that it specifies the role of expectations explicitly through a relatively simplified satellite model.

The main usage of the FRB/US model can be found in the Fed's Tealbook in which all the information dedicated for monetary policy decisions listed. FRB/US has been as the main model producing baseline forecast path of the key macro variables for real GDP growth, unemployment rate, core PCE inflation rate and the federal fund rate coherent to former variables in the model structure. Moreover, simulations on alternative scenarios have been listed such as fiscal restraint, inflation sentiment risks. Based on these simulations, projections for the next five years are provided. In this context, the FRB/US model has served as a key communication tool both within the Federal Reserve and outside of it, as it helps in conveying the range of potential economic outcomes and the uncertainties surrounding them.

It took a considerable time for other central banks to enter upon development of this large scale semi-structural model of their own until the Zero Lower Bound (ZLB) era. After the Great Recession many advanced economies have had hard time dealing with this unprecedented ZLB because their prime tool-mostly linearized Dynamic Stochastic General Equilibrium (DSGE) model-was not ready to be on duty regarding such nonlinearities. On the contrary the FRB/US managed to address this perplexing problem thanks to its flexibility.[2]

It is not surprising that central banks confronting ZLB or acknowledging the need for more flexible and data-oriented model have hopped in the line of developing the FRB/US type model around those days. The Bank of Canada's LENS (Gervais and Gosselin 2014) was the first follower of the FRB/US. It stressed

1) Since its first operation in the field, it has undergone a couple of major revisions until 2018.
2) See Chung et al. (2012), Kiley and Roberts (2017), Bernanke et al. (2019) for the FRB/US model applied to the ZLB topic.

that trend movements could have cyclical implications which conventional setup of DSGE model misses. This is indeed a critical point for small open economies where external trends generate sectoral trends exogenously.

The second wave of FRB/US models went on the floor in 2019. The FR-BDF (Lemoine et al. 2019) of Banque de France and ECB-BASE (Angelini et al. 2019) of European Central Bank share major traits of FRB/US except the former is constrained as its monetary policy is given outside of France. ECB faces asymetric constraints as each member country operates its fiscal policy independently. Nevertheless several contributions were made through the line of the developments.

Ⅲ. Model

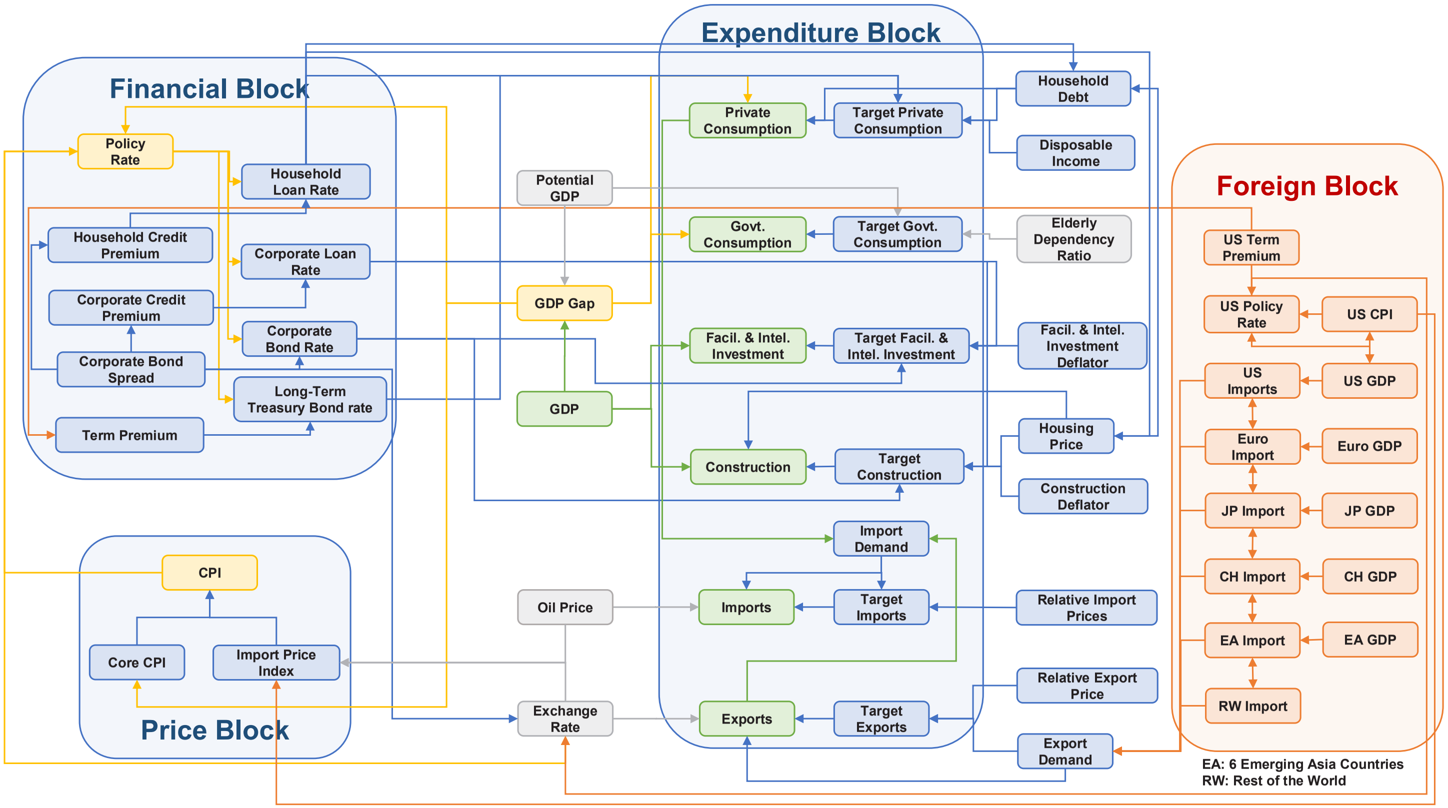
BOK-LOOK is divided into four main blocks: external, expenditure, price, and finance, and consists of 146 endogenous variables and 207 equations as described in Figure 1.

The external block includes key trading partners such as the U.S. and China, and oil prices. It adopts the assumption of block exogeneity, where economic conditions in key trading partners and oil prices unilaterally impact Korea, while Korea's financial and economic conditions do not influence the external block.

The expenditure block consists of consumption, investment, exports and imports, and government consumption. GDP is indirectly backed out as a sum of the expenditure blocks (private consumption, facility investment, construction investment, government consumption, exports, and imports), and the GDP gap is calculated as the difference between GDP and potential GDP. The GDP gap then interacts with the price block which covers core and consumer prices.

Lastly, to incorporate detailed mechanism of monetary policy transmission, the financial block includes not only various types of interest rates with which agents in the model economy face, but also the household debt dynamics to address financial stability issues.

Figure 1. Overview of BOK-LOOK



1. Basic Composition of FRB/US Type Model

The basic composition of most model blocks is divided into long-run and the short-run of the main variable, where the long-run dynamics reflect cointegration based on economic theories while the short-run part of the model admits various kinds of frictions and reduced-form relationships to enhance the empirical fit of the model.

The main idea of the FRB/US type models aims for both theoretical coherence and goodness-of-fit by segregating the role of long-run and short-run part of the model. In the past, it omitted the explict role for the expectation of agents in the model economy. However, it was challenging to explicitly include expectation terms to respond to the Lucas critique as the model grew in size.

To this end, the FRB/US model Brayton and Tinsley (1996) introduced a clever way to adopt the expectation terms into their model in a semi-structural fashion. It starts by formulating an optimization problem for agents of the model economy whose expectation is essential for policy analysis.

Forward-Looking Cost Minimizing Framework

Define a cost minimizing problem for an agent who determines her choice at each period as,

$$\min_{y_t} \sum_{i=0}^{\infty} \beta^i \left[(y_{t+i} - y^*_{t+i})^2 + \sum_{k=1}^{m} b_k \left((1-L)^k y_{t+i} \right)^2 \right] \tag{1}$$

where y_t is her control variable at t and y^*_t is the corresponding target variable which she wants to match with y_t. It can be deemed as an ex-ante optimal choice based on economic theories.

We can split the above minimization problem into two parts. The first part argues that the closer her control variable to its target value, the lower the cost. In contrast, the latter suggests her choice to be stable depending on the sequence of coefficients denoted as $b_k \geq 0$ on the past choices represented with the lag operator L. Note that one must formulate her expectation on the path of target variable when solving this problem at every period.

To understand how it's relatable to the economic problem in the real world, suppose that there is a firm optimizing its plan based on its research about a market. It can easily be an (expected) ideal investment plan for a certain length of horizon. However, there can be some status-quo frictions inducing costs like fixed investment costs. In addition, one must update its investment plan whenever one receives new information in the future. In conclusion, such forward-looking environment can be sufficiently transformed into the above problem and generalized as a typical optimization problem in economics.

Behavioral Equation

The target variable is a latent variable representing theoretical behavior of its associated observable. For instance, it is a counterfactual consumption path based on the permanent income hypothesis. In our model, we label equations with respect to observables in level terms as Behavioral equations as they relate variables with solely from the perspective of economic theories as

$$y_t^* = Ax_t \tag{2}$$

Where x_t is a vector of variables cointegrate with the target variable y_t^* summarized by the coefficient vector A.[3)]

The rationale behind of the behavioral equation is that economic theories regarding timeseries may not provide a better prediction in the short-run, but it is helpful to shape movements in the long-run, cointegrations between macroeconomic variables. Therefore, these equations solely focus on theoretical coherence instead of data-fit.

While several aspects of other components of the model will be addressed in details later, behavioral equations are the gist of the FRB/US type model. First, it builds up the nonstationary properties of the model grounded on estimated potential output. Consequently, it allows the model to generate long-run

3) Going back to the previous consumption example, Suppose y_t^* is a counterfactual consumption path linear to a permanent income of a consumer. Then, the right-hand-side is a policy function for the permanent income. The way such policy function being constructed depends on the theory chosen by modelers including even how expectation is formulated. This will be discussed in the Appendix A.

predictions meaningful and interpretable. Second, it governs the expectation part of the short-run dynamics too as it is the target variable on which agents of the model economy must formulate expectations.

In our model, the target variable is denoted with a superscript star (e.g. z_t^*) so that the star-variables always show up on the left-hand-side of behavioral (long-run) equations.

Polynomial Adjustment Cost Equation

The first order derivative of equation (1) boils down to the following optimality condition,

$$\Delta y_t = a_0(y_{t-1}^* - y_{t-1}) + \sum_{k=1}^{m-1} a_k \Delta y_{t-k} + E_{t-1}\sum_{k=0}^{\infty} d_k \Delta y_{t+k}^* + \epsilon_t \quad (3)$$

This equation is labeled as a Polynomial Adjustment Cost (PAC) as it is a consequence of the embedded time-to-build constraints within the convex costs of adjustment (Kozicki and Tinsley 1999).[4]

It's worth to remark that the PAC equation is nothing but an extended version of an error-correction model augmented with a long-run expectation term which reflects the gradual adjustment process to long-term equilibrium and the preemptive reaction according to the change in expectation. The first term on the right-hand-side corrects the misalignments of the observable from its theoretically coherent value with $a_0 > 0$. The second term is an autoregressive part helps to improve the goodness-of-fit of the model. Lastly, the third term is the new addition from the forward-looking nature of the PAC framework. It allows the optimal choice of today can react to the changes in the expected path of the target variable, discounted by the sequence of coefficients d_k discussed in details later.

The separation strategy of the semi-structure model suits well with this framework. All the economic theory can be freely melted in the model through the latent variable y_t^*. At the same time, the model acknowledges the existence of

4) The PAC is originated from Tinsley and developed throughout the FRB/US model project. See Brayton et al. (2000) for detailed derivations and explantions.

frictions in the real world keeping the realized observable y_t away from its theoretical counterpart at least in short-run.

Expectation Formation on BOK-LOOK

At the heart of the reason for calling FRB/US type models as the modern succession of the traditional simultaneous equation system is that they explicitly address the expectation terms mainly through the PAC equation. Hence, how to induce such expectation form into the model would be the proper question to ask.

The direct answer from today’s macroeconomics is to exploit causal structures of a model microeconomic theories (micro-foundation) by appreciating optimizing behavior of individuals governed by the fundamental (structural) parameters invariant to policy changes, widely known as rational expectation or Model Consistent Expectation (MCE), a conventional way to formulate expectation in most structural macroeconomics model like DSGE model. To elaborate, MCE refers to a rational expectation under full-information both in terms of observables and structures of the model so that resulting expectations are consistent with the solution of the model.[5)]

Unfortunately due to the scale of the model, it is challenging to incorporate with fully rational agents who understand and respond to every state variable without making such behavior too sensitive. Besides, unlike a small or medium scale structural models, it is less convincing for an agent of a semi-structural model economy to form her expectation in the semi-structural way per se.

Brayton and Tinsley (1996) suggested another approach by adopting a VAR based Expectation (VARE), formulating expectations from a satellite VAR model to mimic its forward-looking nature in macroeconomics partly allowed to deviate from full-information, rational expectation framework like MCE.[6)]

One can interpret the expectation formulated by economic agents in the VARE environment whose information set is limited (e.g. only observe some of

5) Under MCE, agents of a model economy make their expectations equivalent to the one from full model. Hence the literature also describes MCE as “perfect foresight” and “rational expectation”. See Brayton et al. (1997), Bernanke et al. (2019), Lemoine et al. (2019), Angelini et al. (2019) for further discussions.

6) In Appendix A, we describe satellite VAR models for expectation formation in details.

dataset) or who understand the comovement of observables based on historical patterns only. Hence, modelers can set to what extent how well the VARE approximates the MCE by the intensity of the VAR model settled in. In fact, existing models set the benchmark type of expectations as a hybrid version, where agents in financial sector are equipped with MCE but others are with VARE.

2. Estimation

While significant advances have been made in econometric methods, the large size of FRB/US-type models imposes constraints on estimating them as a single, unified system. Instead, the BOK-LOOK model is estimated in a blockwise fashion, which introduces potential endogeneity issues. These challenges are a recognized trade-off when dealing with models of this scale and complexity.[7)]

The equations in the model were primarily estimated using Bayesian methods, reflecting the limited availability of data in Korea compared to larger advanced economies.[8)] However, for the behavioral equations, many parameters were calibrated without direct data inference. The rationale for indirect estimations lies in respecting the theoretical underpinnings of the cost-minimizing problem. In this framework, an agent's objective is to track the target variable while minimizing unnecessary variability in their choices. However, this problem becomes trivial if the target variable itself does not exhibit substantial variation. In such cases, there is no meaningful trade-off between the dual objectives of tracking the target and maintaining stable choices.

This consideration highlights a critical feature of the PAC framework: it is most effective when the target variable exhibits greater variability than the control variable, thus providing error-correcting driven by theories. This perspective contrasts with conventional protocols, where target or latent variables are often modeled as highly persistent or stable processes (e.g. potential output). Consequently, estimating target variables in the usual least-squares fashion may

7) We setup a satellite DSGE type model regarding to inflation estimation (Phillips equation) due to the concern of severe endogeneity, in line with FRB/US and ECB-BASE.

8) Most of the equations are estimated with data span from 2000Q1 to 2024Q2.

be misleading, as fitted values tend to exhibit less variability than their regressors, resulting in positive R^2 but less informative dynamics.

To address these limitations, we placed a strong emphasis on calibration and the use of more informative Bayesian priors targeting theoretical moments rather than empirical ones. This approach ensures that the model adhered to the theoretical motivations underpinning its structure.

For the core satellite VAR models, estimation followed a conventional Bayesian VAR framework with the inclusion of a typical Minnesota prior (Doan et al. 1984). Block exogeneity was assumed when the model was augmented with block-specific variables, preserving the structural integrity of the external components.

The short-run PAC equations were also estimated using Bayesian methods. Once priors for the coefficients in the PAC equations were set, the long-run expectation terms were computed alongside estimates from the associated satellite VAR models. The likelihood of each equation was then evaluated using a Markov Chain Monte Carlo (MCMC) simulation.

To ensure robust estimation, Gibbs sampling was applied for the satellite VAR models, with a total of 500,000 iterations. For the PAC equations, the Metropolis-Hastings algorithm was employed, iterating 1,000,000 times. These extensive iterations reflect the complexity of the estimation process and the need for precision in aligning the model's outcomes with its theoretical and empirical objectives.

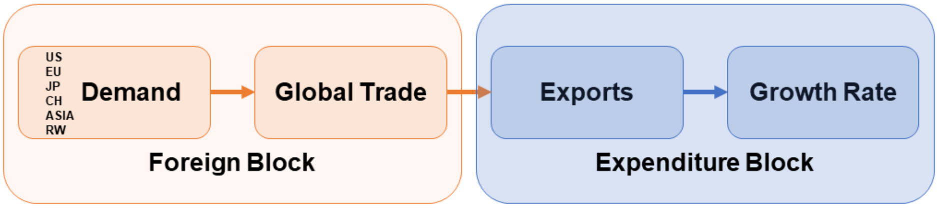
3. External

China/EU/Japan/Asia/Rest of the World

Enhancing the resolution of foreign blocks represents a major contribution to the literature of the FRB/US type models. Considering the essential role of the exports in the Korean economic growth, we judge that it would be reasonable to put significant efforts into this part.

We divided the global economy into six blocks; U.S., China, EU, Japan, rest-of-Asia and rest-of-the-world regarding Korea's export value chain in terms of the real trade variables only, except the U.S. block discussed later. The key is to refine the chain reaction impact of global trade on Korean export. For example, suppose a positive demand shock to one of the block economies. Initially, the block economy increases demand more product from Korea directly, which can be labeled as the first order impact. On top of that, the increased demand would not only increase demand from a single country but to other economies as well. It can be interpreted as a spillover demand shock on other blocks and eventually increase the import demand for Korean goods, the second order impact, summarized as in Figure 2.

Figure 2. Global Trade Channel



To translate the idea into the model, we interwined each block in terms of gaps (cycles) as;

$$\widehat{M}_t^j = \rho_M^j \widehat{M}_{t-1}^j + \beta_M^j \hat{y}_t^j + u_{M,t}^j, \;\; j \in \{\text{six blocks}\} \tag{4}$$

where $\widehat{M}_t^j$ is an import gap of global block j affected by its past term and current output gap $\hat{y}_t^j$.

A key consideration was to account for how we would operate the model in the forecast process when designing the external parts. The idea was that we would be given the expected output growth path of the global economies which led us to set the streamline as follows;[9] First, we derive the output gap of each global block as,

9) Instead of letting the model to fully endogenizing each foreign block's dynamics of GDP and import demand, we designed the model to rely on the bank's international division's forecast on growth rates of GDP of the global blocks.

$$\hat{y}_t^j = \rho_y^j \hat{y}_{t-1}^j + \beta_y^j X_{D,t}^j - \omega_j \hat{P}_t^{oil} + u_{y,t}^j \tag{5}$$

$$X_{D,t}^j = \sum_{i \neq j} \gamma_{i,j} \widehat{M}_t^j \tag{6}$$

where $X_{D,t}^j$ is a spillover demand for country j from as a sum of the rest of global blocks in the model. We also allowed oil price gap to negatively correlated to the output gap of each economy so that the oil price shock can act as a global demand shock.

The model system then jointly determines the output gap and the import gap of the global economy to compute the aggregate demand for Korean export which will be discussed in a later chapter.

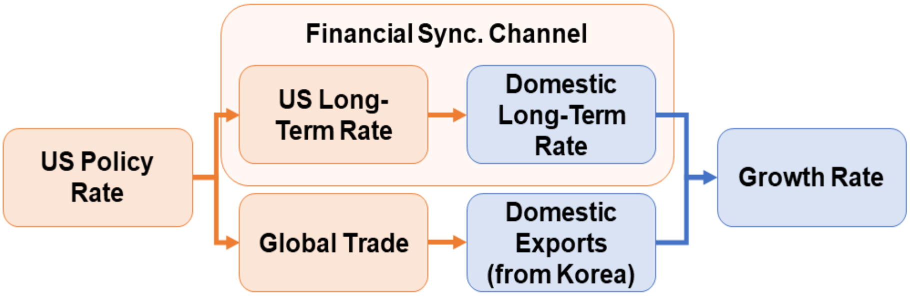
U.S.

While modeling the foreign parts in real economy terms only, it was necessary for us to come up with a way to implement the global financial market impact to the domestic economy. Such significance is clearly demonstrated in Figure 3. First, U.S. monetary policy is crucial in that it affects the long-term interest rate of U.S. which tightly engaged with that of Korea through the term-premiums labeled as a financial synchronizing channel. Secondly, U.S. monetary policy also runs into the real sector of global economy via U.S. output gap channel, the bottom part of the Figure 3.

To this end, we applied a special treatment to the U.S. block by extending it as a version of small scale New Keynesian model modified for empirical fit (Carabenciov et al. 2008).[10)]

10) The IMF's QPM model (Carabenciov et al. 2008) is a semi-structural macroeconomic model linking key variables such as output, inflation, exchange rates, and interest rates. The model incorporates gap-based frameworks, including Phillips curves for inflation and is curves for output, and features forward- looking components to capture expectations. Additionally, it also incorporates backward terms in its equations without their microfoundations, to ensure the empirical flexibility.

Figure 3. U.S. Financial Channel



4. Expenditures

Private Consumption

The BOK-LOOK model's target private consumption is basically developed using the permanent income hypothesis. This approach seeks to model consumption based on a theoretically sound variable that represents the permanent income path, with the constraint that this path aligns closely with observed private consumption.[11)][12)] On top of it, the model assumes that household debt negatively influences private consumption in the long run (Lombardi et al. 2017). However, to avoid excessive sensitivity to recent high debt levels, the debt coefficient in the model's forecasting framework was moderated. This adjustment helped prevent recent debt from excessively dampening consumption compared to historical averages.

$$\ln C_t^* = \beta_{c,0} + \ln Y_t^{income} + \beta_{c,1} \ln Debt_t + \beta_{c,2} Dummy_{13Q1,19Q4} + \beta_{c,3} Dummy_{\overline{20Q1}} + \epsilon_{c,t} \tag{7}$$

11) As real permanent income data at a quarterly frequency is unavailable, an indirect method was employed to approximate permanent income

$$Y_t^{income} = Potential\ GDP_t \times \frac{Disposable\ income_t}{Nominal\ GDP_t}$$

Here, the final term in the equation represents the ratio of nominal disposable income to nominal GDP. This approach captures a stable, long-run income level that adjusts for deviations in disposable income.

12) Initial efforts included constructing a satellite VAR model that incorporated current national income to generate an income expectation proxy. However, this only differed in cyclical components from the current income path, indicating that the observed income path was already effectively representing the long-term trend.

The above is the behavioral equation for the target private consumption C_t^* consists of permanent income Y_t^{income}, household debt $Debt_t$, and dummy variables for correcting temporary trend changes caused by the Great Recession and Covid-19.[13)]

The short-run equation for private consumption growth $\Delta \ln C_t$ is based on a polynomial adjustment cost framework, similar to that used in the FRB/US model.

$$\Delta \ln C_t = \alpha_{c,0}(\ln C_{t-1}^* - \ln C_{t-1}) + \alpha_{c,1}\Delta \ln C_{t-1} + E_t\left[\sum_{k=1}^{\infty} d_k \ln \Delta C_{t+k}^*\right] \\ + \gamma_{c,1}\Delta \hat{y}_t + \gamma_{c,2} i_{HH,t} + \gamma_{c,3}\Delta \ln Debt_t + \gamma_{c,4}\Delta_4 \ln PP_t + u_{C,t} \tag{8}$$

This setup is augmented with additional terms, including output gap $\hat{y}_t$, household interest rates $i_{HH,t}$, debt growth $\Delta \ln Debt_t$, and real purchasing power PP_t. Real purchasing power, in turn, was a judgment estimated by exogenous variable derived from nominal wage and aggregate employment projections which helps improving the forecast accuracy on consumption paths. Following Gervais and Gosselin (2014), Angelina et al. (2019) output gap in the consumption dynamics represents the hand-to-mouth agents of the economy. In contrast to the long-run behavior, we set the debt growth boosts the consumption growth in the short-run in lines with Lombardi et al. (2017).

Facilities and Intellectual Property INVESTMENT

We decomposed the aggregate investment into facilities & intellectual property (henceforth FI) and construction part, given that the former was highly exposed to global economic conditions while the latter was dependent to the domestic environment mostly. Nevertheless, both follow the canonical specification from Gervais and Gosselin (2014), Angelina et al. (2019) where the behavioral equations are derived from the first order condition based on the neoclassical production function as follows,[14)]

13) A dummy variable $Dummy_{\bar{\tau}}$ has a value 0 for $t < \tau$ and 1 for $t \geq 1$. For a temporary dummy variable $Dummy_{\tau_1,\tau_2}$ assigns value 1 for $\tau_1 \leq t \leq \tau_2$ and 0 otherwise.

14) Lemoine et al. (2019) also derived the long–run equation for business investment in similar fashion, but they generalized it with the CES production function.

$$\ln I_t^* = \beta_{I,0} + \beta_{I,1} \ln \overline{Y_t} + \beta_{I,2} Dummy_{\overline{20Q2}} - UC_{I,t} \tag{9}$$

where the target FI investment I_t^* is linked with the potential output $\overline{Y_t}$ in the long-run. Originally, we constrained the target variable to cointegrates with the potential output, but the trend of the FI investment has grown a bit faster than that of output in Korea and we reflected it into the behavioral equation.[15] Additionally, we adopted a Covid-19 dummy to correct the trend of the FI investment which had recently departed from that of the GDP described in the equation (9).

The user cost of capital $UC_{I,t}$ represents the marginal cost of capital from the empirical perspective.

$$UC_{I,t} = (i_{Firm,t} + i_{CB,t})/2 - \pi_{cpi,t}^{yoy}/4 + \delta_{I,t} \tag{10}$$

Though there exists a heterogeneity in firms financing costs in terms of the different borrowing rate they face, we averaged the cost with the firm's loan rate $i_{Firm,t}$ and corporate fund rate $i_{CB,t}$ for parsimony and then denominated it with consumer price index inflation. Lastly, we also take the depreciation rate for FI investment goods $\delta_{I,t}$ into account for measuring the user cost of capital.

The short-run dynamics of FI investment I_t is constructed under the polynomial adjustment cost framework,

$$\begin{aligned} \Delta \ln I_t = {} & \alpha_{I,0}(\ln I_{t-1}^* - \ln I_{t-1}) + \alpha_{I,1} \Delta \ln I_{t-1} + E_t \left[\sum_{k=1}^{\infty} d_k \ln \Delta I_{t+k}^* \right] \\ & + \gamma_{I,1} \Delta \hat{y}_t + \gamma_{I,2} \Delta \ln P_{I,t} + \gamma_{I,3} \ln DRAM_t + u_{I,t} \end{aligned} \tag{11}$$

The output gap term $\hat{y}_t$ takes part in to explain the short-run volatility of the FI investment. On top of that we tightly linked it to the variables relevant to the global business cycle as well. One of these variables is changes in the FI

15) Gervais and Gosselin (2014) addressed this issue by introducing a variable representing the share of mechiery & equipment and construction. However, Korea's ratio of aggregate investment to output has shown a slightly downard trend. This is mainly due to the unique characteristics inherits from the construction investment in Korea which will be dicussed in the later chapter of this paper with more details.

investment deflator $P_{I,t}$, which strongly correlates with the nominal exchange rate changes (79% in sample period). The other is the excess demand index of semi-conductor from Gartner $DRAM_t$ reflecting the prominent role of the semi-conductor industry in Korean business cycle.

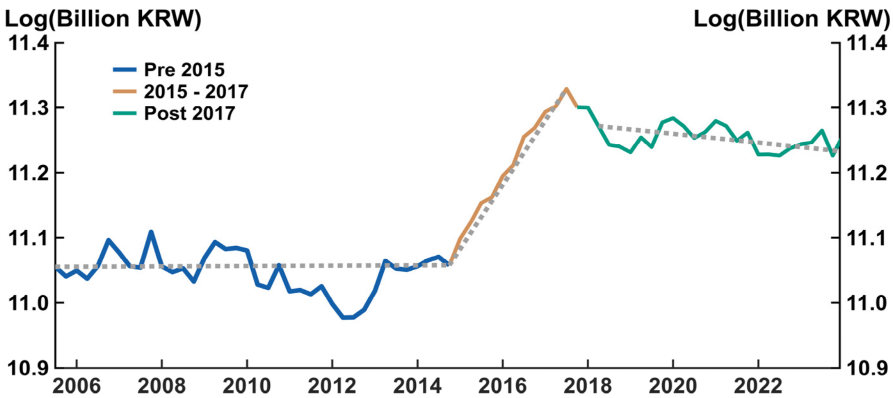
CONSTRUCTION INVESTMENT

The behavioral equation for the construction investment target IH_t^* is designed in similar fashion with the FI investment equation in which the dependent variable comoves with the potential output $\overline{Y}_t$,

$$\ln IH_t^* = \beta_{IH,0} + \beta_{IH,1} \ln \overline{Y}_t + \beta_{IH,2} \ln GB_t - UC_{IH,t} \tag{12}$$

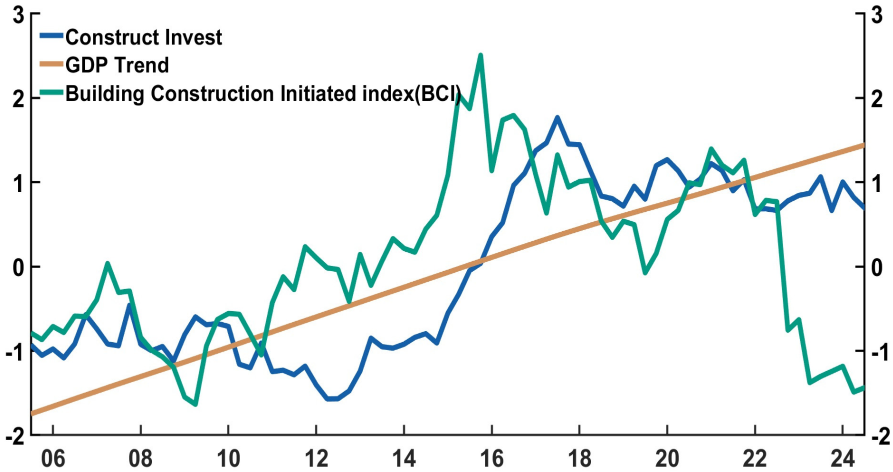
Explaining the rest of the regressors follows; Capturing construction investment was one of many challenges throughout this project due to its peculiar movement in trend. The left panel of Figure 4 illustrates such difficulty is mainly resulting from the construction boom started in 2014 and peaked around 2017.

Figure 4. Construction Investment Plot



Note: Each solid line depicts log of construction investment with different time periods. Each dotted line simply draws linear trends for associated subsamples.
Data source: ECOS

Figure 5. Construction Investment and Its Regressors



Note: Each line depicts log of construction investment (Blue), HP-filtered trend GDP (Golden), and log of Building Construction Initiated index (Green). All variables are normalized by its mean.
Data source: ECOS

These sudden changes are never easy to reflect in a structural manner because it is unlikely a macroeconomic consequence. Instead, the consensus identifies its main driver rather as the pivotal of government stance on the housing market. To reconcile this, we applied an index variable GB_t which quantifies the area where construction began as one can see the comovement from the Figure 5.

$$UC_{IH,t} = (i_{Firm,t} + i_{CB,t})/2 + \pi^{yoy}_{cpi,t}/4 + \delta_{IH,t} \tag{13}$$

Another characteristics of construction investment beside its aforementioned jump is that unlike the other GDP components, it does not show a growth in level. Arguably the Figure 5 plainly depicts the growth in level solely comes from the period of sudden jump. It puzzles the cointegration relationship between the construction investment and the GDP. Moreover, it is reasonable to predict that this gap will widen further in the out-of-sample period. From timeseries aspect, such gap is worrisome unless we can interpret it in terms of well-founded economic theories forcing the construction investment to grow in the future.

While the long-run behavior of the construction investment calls for some further measures to take on, short-run dynamics are based on the PAC form.

$$\begin{aligned}\Delta \ln IH_t = {} & \alpha_{IH,0}(\ln IH^*_{t-1} - \ln IH_{t-1}) + \alpha_{IH,1}\Delta \ln IH_{t-1} \\ & + E_t\left[\sum_{k=1}^{\infty} d_k \ln \Delta IH^*_{t+k}\right] + \gamma_{IH,1}\Delta \hat{y}_t + \gamma_{IH,2}\Delta \ln P_{IH,t} \\ & + \gamma_{IH,3}\Delta \ln HPI_t + \gamma_{IH,4}\Delta_4 \ln BCI_t + u_{IH,t}\end{aligned} \tag{14}$$

Augmented variables are changes in output gap $\hat{y}_t$, deflator of construction investment $P_{IH,t}$, housing price index HPI_t and annual growth rate of Building-Construction-Initiated BCI_t in sequence. When determining the specification of the construction investment part, we examined which of the housing price and building-construction-initiated should be included in the behavioral and PAC equation respectively. We determined to put housing price into the short-run equation rather than into the long-run one because housing price wasn't quite successful in explaining the previous gap.

GOVERNMENT CONSUMPTION

While the public sector takes a certain part of the Korean macroeconomy, we took a concise approach to describe it.[16] The target government consumption was explicitly modeled as a function of potential GDP and an aging index,

$$\ln G^*_t = \beta_{G,0} + \beta_{G,1} \ln \overline{Y}_t + \beta_{G,2} EDEPR_t \tag{15}$$

where the $\overline{Y}_t$ is the potential output gap and $EDEPR_t$ is the elderly-rate index. The last term is included to shed light on the incoming demand from the social security perspective.[17]

The growth rate of government consumption was a function of the GDP gap interpreted reflecting its response to the business cycle.

$$\Delta \ln G_t = c_G + \alpha_{G,0}(\ln G^*_{t-1} - \ln G_{t-1}) + \alpha_{G,1}\hat{y}_t + u_{G,t} \tag{16}$$

Interestingly, simple error-correction estimation favors the countercyclical fiscal behavior, implying that the sustainable government finance is considered.

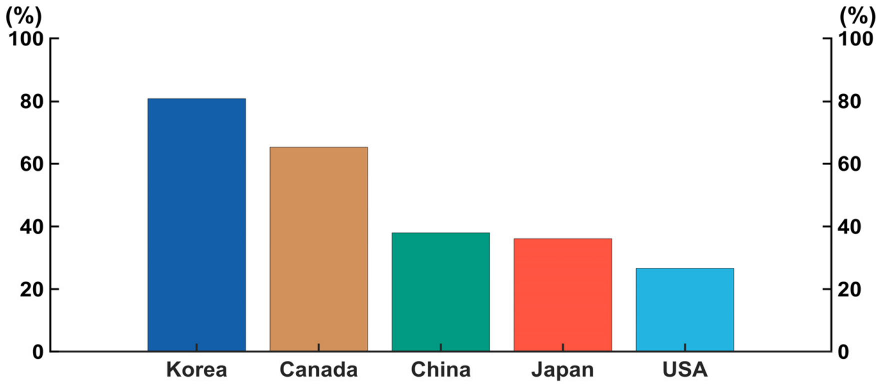
16) We are planning to elaborate on the fiscal part by carefully introducing the government budget constraint and setting up each variables−spending, transfer, taxation and debt operation− in it.

17) We add extra dummy variable for forecasting practice after the Covid−19.

EXPORT

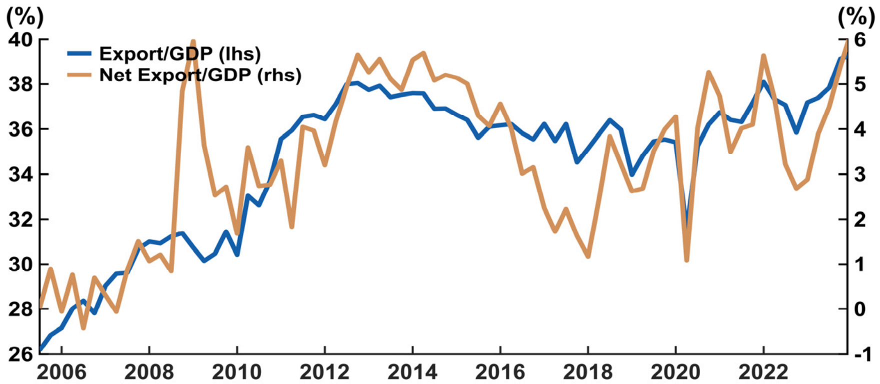
Korea is one of the most trade-dependent countries in the world. Figure 6 compares the trade dependency among relevant countries and reveals this tendency solidified after the pandemic era. Figure 7 also asserts that the export share in GDP is growing both in gross export and net export. In addition, Figure 7 captures a structural change at the break of Covid-19 where the correlation between Korea's GDP and export rose drastically, implying the export has become the main driver of the GDP growth.

Figure 6. Trade Percentage of GDP



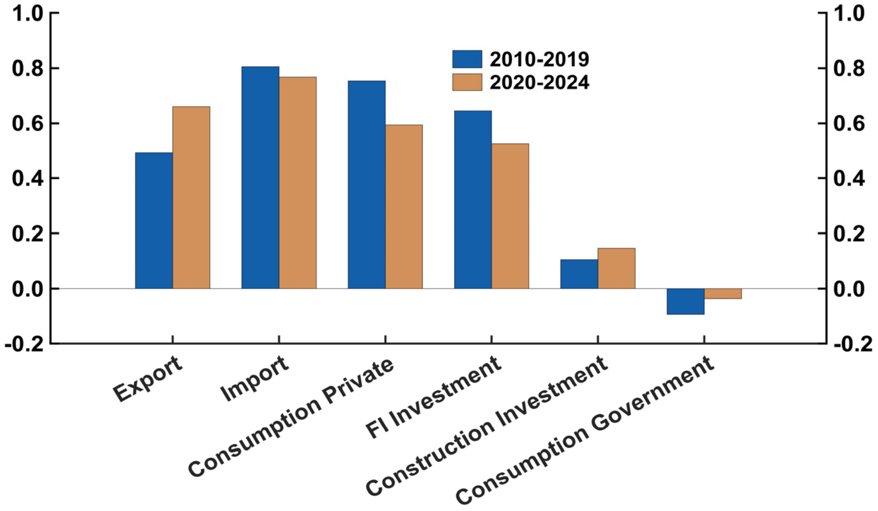
Note: Each bar depicts trade-to-GDP ratio of each country where the trade is defined as the absolute sum of exports and imports
Data source: OECD

Figure 7. Trade Share in GDP



Note: Each line depicts export divided by GDP (Blue, Left axis), and net export divided by GDP (Golden, Right axis)
Data source: ECOS

Figure 8. GDP Componentsí Correlation with GDP QoQ Growth



Note: Blue bars show before Covid-19 and golden bars do after Covid-19
Data source: ECOS

This finding highlighting the significance of export assigns a task for modeling Korean macroeconomy. The key is to figure out the export demand of Korea as an export of a country mirrors its counterparts import demand. Hence our target export X_t^* was modeled as a function of export demand X_t^{demand}, defined as the weighted average of import demand from major trading partners M_t^j which will be discussed later[18]:

$$\ln X_t^* = \beta_{X,0} + \beta_{X,1} \ln X_t^{demand} \tag{17}$$

$$\begin{aligned} X_t^{demand} = & \zeta_{US}^X M_t^{US} + \zeta_{EU}^X M_t^{EU} + \zeta_{CH}^X M_t^{CH} \\ & + \zeta_{EA}^X M_t^{EA} + \zeta_{JP}^X M_t^{JP} + \zeta_{RW}^X M_t^{RW} \end{aligned} \tag{18}$$

The growth rate of exports is driven by the growth rate of export demand and the nominal KRW/USD exchange rate following the standard given by the benchmark models (Brayton and Tinsley 1996, Gervais and Gosselin 2014, Lemoine et al. 2019, Angelini et al. 2019),

18) The weight of each trading partner is calculated using bilateral trade data from the IMF DOT database.

$$\begin{aligned}\Delta \ln X_t = & \ c_X + \alpha_{X,0}(\ln X^*_{t-1} - \ln X_{t-1}) + \alpha_{X,1}\Delta \ln X^{demand}_t \\ & + \alpha_{X,2}\Delta_4 \ln \mathrm{EXR}_t + u_{X,t}\end{aligned} \tag{19}$$

where EXR_t is the nominal exchange rate.

IMPORT

The series of works in the FRB/US type model, the backbone of the target imports M^*_t is the domestic import demand M^{demand}_t, derived from total import demand across domestic expenditure components[19]:

$$\ln M^*_t = \beta_{M,0} + \beta_{M,1} \ln M^{demand}_t \tag{20}$$

$$M^{demand}_t = \zeta^M_C C_t + \zeta^M_I I_t + \zeta^M_{IH} IH_t + \zeta^M_G G_t + \zeta^M_X X_t \tag{21}$$

The share of import demand on each component was calibrated to target the sample average of the import inducement coefficient from 2000 to 2024.

The growth rate of imports is similarly driven by export demand growth and the nominal KRW/USD exchange rate:

$$\begin{aligned}\Delta \ln M_t = & \ c_M + \alpha_{M,0}(\ln M^*_{t-1} - \ln M_{t-1}) + \alpha_{M,1}\Delta \ln M^{demand}_t \\ & + \alpha_{M,2}\Delta \ln \mathrm{EXR}_t + u_{M,t}\end{aligned} \tag{22}$$

While it is more likely that the import price would act directly as determinant of imports rather than exchange rate, there are at least two reasons behind. First, not only the empirically, the timeseries of exchange rate change (-0.58) has been correlated higher than that of import price (0.23), the direction of the latter case is puzzling as well. The other reason regards to the global value chain. The main structure of Korean economy's trade can be characterized by importing primary intermediate good and exporting secondary intermediate good. Therefore, positive correlation between the import inflation and the growth rate of import

19) Gervais and Gosselin (2014) disaggregated import volumes into commodity and non-commodity to reflect the trade structure of Canada. It results in augmenting real effective exchange rate or real commodity price in the target imports equation as well.

that might seem counterfactual can be seen differently depending on how the value chain is affected by the exchange rate and import price. However, since this disaggregation is left as a future task, the model was constructed as described above.

5. Prices

CONSUMER PRICE INDEX(FOOD AND ENERGY PRICE EXCLUDED)

The gist of the monetary policy operation lies in the (short-run) linkage between the real economy's activity and nominal inflation, denoted as Phillips equation. Since it entails significant endogeneity by nature, most of the FRB/US type models have estimated this part by setting up a satellite model for price in terms of dynamic stochastic general equilibrium fashion (Brayton and Tinsley 1996, Gervais and Gosselin 2014, Lemoine et al. 2019, Angelini et al. 2019).

Brayton (2013) documented the satellite model for FRB/US model's price and wage sector where it is the core inflation that is determined through New Keynesian Phillips equation instead of headline inflation.[20] In lines with the high exposure to global factors governing food and energy prices, we also replaced consumer price index (CPI) inflation π_t with consumer price index excluding food and energy price (Core CPI) inflation $\pi_{Core,t}$ in our Phillips equation and let the CPI inflation be computed by combining the domestic inflation with the import price inflation.

$$\pi_{Core,t} = (1-\phi_1-\phi_2)\overline{\pi}_t + \phi_1 \pi_{Core,t-1} + \phi_2 E_t \pi_{Core,t+1} + \phi_3 \hat{y}_t + u_{Core,t} \tag{23}$$

The Phillips equation in most of the FRB/US type models augment an inflation attractor variable denoted as $\overline{\pi}_t$. It represents the long-run inflation expectation.[21] It allows us to say something about (de)anchoring of private

20) Gervais and Gosselin (2014) also used a modified version of core inflation that excludes the prices of motor vehicles and regulated services (CPIX inflation) to clear out variations suspected to be irrelevant to the relationship between underlying inflation and the output gap. This is also pointed out in Brayton (2013) where he argued that the core inflation better aligns the sector's main price measure with sticky price spirit of the New Keynesian Phillips Curve.

sector's long-run expectation to the inflation target of the central bank.

$$\overline{\pi}_t = (1-\delta_1)\frac{\pi^*}{4} + \delta_1 \overline{\pi}_{t-1} + \delta_2(\pi_{Core,t-1} - \overline{\pi}_{t-1}) + u_{\overline{\pi},t} \tag{24}$$

The long-run expectation on inflation is formulated through adaptive expectation based on the past core inflation and the fixed inflation target π^*.

CONSUMER PRICE INDEX(CPI) INFLATION

Since the core inflation has replaced the role of CPI inflation in the Phillips equation, the target CPI $P^*_{cpi,t}$ is determined in a simple accounting fashion as a combination of the core CPI $P_{core,t}$ with the import price.[22)]

$$\ln P^*_{cpi,t} = \nu_{cpi} \ln P_{Core,t} + (1-\nu_{cpi}) \ln P_{M,t} \tag{25}$$

where $P_{M,t}$ denotes import price index.

Given the cointegration of the nominal variables, CPI inflation is estimated via a conventional error-correction model, allowing both short-run and long-run deviations from the structure,

$$\begin{aligned}\pi_{cpi,t} = c_{cpi} &+ \alpha_{cpi,0}(\ln P^*_{cpi,t-1} - \ln P_{cpi,t-1}) + \alpha_{cpi,1}\pi_{Core,t} \\ &+ \alpha_{cpi,2}\Delta \ln P_{M,t} + u_{cpi,t}\end{aligned} \tag{26}$$

Estimates of coefficients seem to imply the effect of import price part is insignificant compared to that of core CPI. However, one must note that the variability of the former is much larger than that of latter.

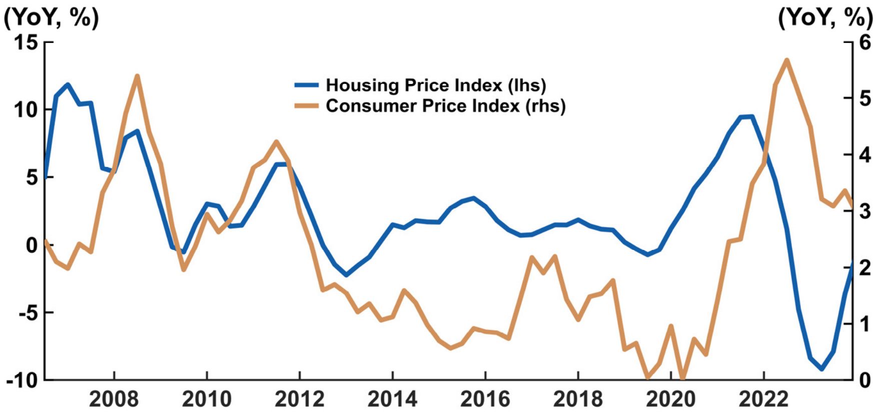
21) $\overline{\pi}_t$ is equivalent to the shifting endpoint variable for inflation in the satellite VAR model. It represents the long-run expectation on inflation indicating the possibility of inconsistency between expectations of short-run and the long-run.

22) Coibion and Gorodnichenko (2015) applied both CPI and GDP deflator as a measurement of U.S. inflation in Phillips equation and found the result was robust regardless of which data they took. However, most small open economies are facing huge dispersion between CPI and GDP deflator mainly due to the import price deflator (Huh 2024). We would like to stress out the urgent need for investigating the potential consequence of taking CPI into the Phillips equation for a small open economy.

HOUSING PRICE

Housing price has drawn attention for both monetary and fiscal policy authorities since it has been considered as a main driver of the growing level of household debt. Indeed, such speculations have made the housing price more volatile as shown in Figure 9. Although the housing price is not directly included in the basket of consumer price index, it can stimulate the aggregate demand by changing housing asset values. In particular, Korean economy observed a surge of housing price boom during the pandemic era which delivered heterogenous effects to home owners and tenants.

Figure 9. CPI and Housing Price Inflation



Note: Each line depicts Housing price (Blue, Left-axis) and CPI (Golden, Right-axis) on Year-on-year basis.
Data source: ECOS

However, accounting for the macroeconomic effect to housing price is never a easy task because it has always tightly engaged with the regulations from political perspectives hardly generalizable. Hence we modeled rather in a parsimonious manner that the housing price comoves with the price level and the market rate for the household borrowing,

$$\ln HPI_t^* = \beta_{hpi,0} + \beta_{hpi,1} \ln P_{cpi,t} + \beta_{hpi,2} i_{HH,t} \tag{27}$$

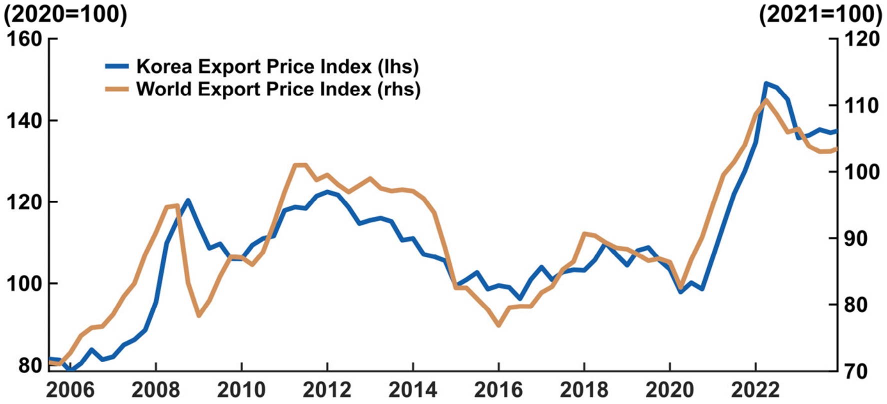
$$\begin{aligned} \Delta \ln HPI_t = c_{hpi} + \alpha_{hpi,0}(\ln HPI_{t-1}^* - \ln HPI_{t-1}) \\ + \alpha_{hpi,1} \Delta \ln HPI_{t-1} + \alpha_{hpi,2} i_{HH,t} + u_{hpi,t} \end{aligned} \tag{28}$$

where $P_{cpi,t}$ is a CPI level and $i_{HH,t}$ is a borrowing rate for households.

TRADE PRICES

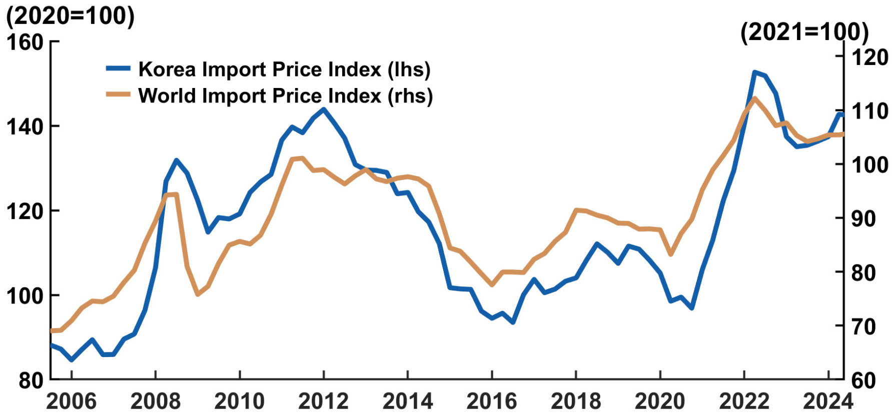
Both export and import prices are coupled with each's corresponding price accounting for market competitiveness the Korean production faces as shown in Figure 10 and Figure 11. Interestingly, it is prominent that the world prices were followed by the Korea's prices during the Great Recession and Covid-19 periods, although one can observe some dispersions arising recently.

Figure 10. Export Prices: Korea and World



Note: Each line depicts export price index of Korea (Blue, Left-axis) and export price index of world (Golden, Right-axis)
Data source: ECOS and CPB

Figure 11. Import Prices: Korea and World



Note: Each line depicts import price index of Korea (Blue, Left-axis) and import price index of world (Golden, Right-axis)
Data source: ECOS and CPB

Even so, we simplified the specification of the trade prices, which helped clarify the causal chain from the world prices to the others.

Evidently the mechanisms of pricing of export and import price are different. We emphasized the price-taking behavior on the export target $P^*_{X,t}$ focusing on the market competitiveness discussed above.

$$\ln P^*_{X,t} = \beta_{px,0} + \beta_{px,1} \ln P^{WD}_{X,t} \tag{29}$$

$$\begin{aligned} \Delta \ln P_{X,t} = {} & c_{px} + \alpha_{px,0}(\ln P^*_{X,t-1} - \ln P_{X,t-1}) + \alpha_{px,1} \Delta \ln P_{X,t-1} \\ & + \alpha_{px,2} \Delta \ln P^{WD}_{X,t} + u_{px,t} \end{aligned} \tag{30}$$

where $P^{WD}_{X,t}$ denotes the world export price index.

One the other hand, recalling the value chain regarding to the trade-importing primary intermediate good-international oil price and the dominant currency paradigm must be considered for the import price structure,

$$\ln P^*_{M,t} = \beta_{pm,0} + \beta_{pm,1} \ln EXR_t + \beta_{pm,2} \ln P_{oil,t} \tag{31}$$

$$\begin{aligned} \Delta \ln P_{M,t} = {} & c_{pm} + \alpha_{pm,0}(\ln P^*_{M,t-1} - \ln P_{M,t-1}) + \alpha_{pm,1} \Delta \ln EXR_t \\ & + \alpha_{pm,2} \Delta \ln P_{oil,t} + u_{pm,t} \end{aligned} \tag{32}$$

EXCHANGE RATES

The current version of the model accomodates the KRW/USD as a single exchange rate. Basically, nominal exchange rate EXR_t is determined by the uncovered interest rate parity with given steady-state $\ln EXR_{ss}$,

$$\begin{aligned} \ln EXR_t = {} & UIP_t + (1 - \alpha_{EXR,0}) \ln EXR_{ss} + \alpha_{EXR,0} \ln EXR_{t-1} \\ & + \alpha_{EXR,2}(i_{US,t} - i_t) + u_{EXR,t} \end{aligned} \tag{33}$$

where UIP_t is a risk premium term that captures short-run fluctuation in terms of the trade surplus,[23)]

$$UIP_t = c_{UIP}(\ln X_t - \ln M_t) \tag{34}$$

23) We also operate a modified version of the risk premium where the 3-year corporate bond rate is included to utilize their empirical fit to each other.

6. Financial Market

MONETARY POLICY

Monetary policy authority in the model determines the call rate i_t. Ultimately, what we want to implement would definitely be simulating various types of policy rules for policy analysis. Currently, we set the baseline monetary policy rule to follow the standard Taylor rule reflecting interest rate smoothing adopted by many other models of central banks.

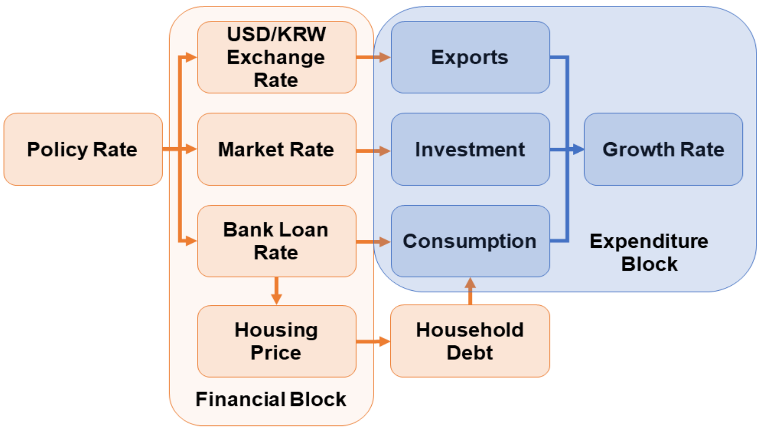
$$i_t = \phi_i i_{t-1} + (1-\phi_i)(\pi^* + r^* + \phi_\pi(\pi_t^{yoy} - \pi^*) + \phi_y \hat{y}_t) + u_{i,t} \tag{35}$$

Note that we defined r^* as an annual real equilibrium rate assumed as a constant.[24)]

TREASURY BOND RATES

The main mechanism of the monetary policy transmission of the BOK-LOOK takes the route from the policy rate to each market rate confronted by agents of the model economy desribed as in Figure 12.

Figure 12. Monetary Policy Transmission Channel



24) In the latest version of FRB/US model, the equilibrium value of the real policy rate is implicit in the model's structure and it permits the intercept of various monetary policy reaction functions to adapt in sumulations to persistent movements in the simulated level of the implicit equilibrium rate. In the BOK-LOOK, we used the mean of the estimate from Laubach and Williams (2003) for Korean economy.

To implement such spillovers it is necessary for the model to give shape to a term structure of the bond rate as market rates have different maturities by their nature.

Following Angelini et al. (2019) we introduced three and ten-year treasury bond rate to configure the term structure. Bond rates are divided into the expected term of short-run interest rate and the term premium as its remainder,

$$i_{TB10Y,t} = E_t \sum_{k=0}^{39} i_{t+k} + \eta_{TB10Y,t} \tag{36}$$

$$i_{TB3Y,t} = E_t \sum_{k=0}^{11} i_{t+k} + \eta_{TB3Y,t} \tag{37}$$

where $\eta_{TB10Y,t}$ and η_{TB3Y} is denoted as the term premium of ten and three-year respectively.[25)]

MARKET RATES

There are three market rates in the model; three-year corporate bond rate (AA-), bank loan rate to firm and household. First, the three-year corporate bond rate $i_{CB,t}$ simply tied with the treasury bond rate of the same maturity $i_{TB3Y,t}$ and their spread $\eta_{CB,t}$,

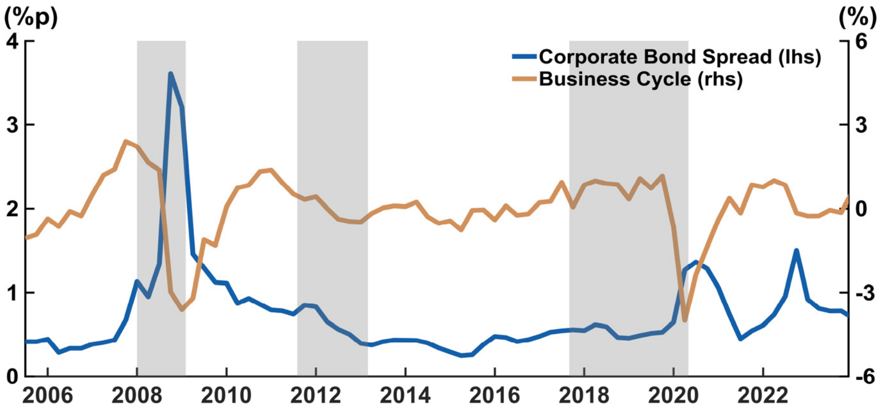
$$i_{CB,t} = i_{TB3Y,t} + \eta_{CB,t} \tag{38}$$

$$\eta_{CB,t} = (1-\rho_{CB})\bar{\eta}_{CB} + \rho_{CB}\eta_{CB,t-1} + \alpha_{CB}\hat{y}_{t-1} + u_{CB,t} \tag{39}$$

Figure 13 supports the idea of having the business cycle term in the spread equation, although their comovement shows a bit of state dependency but it outstands in the time of crisis. Hence this specification stands for representing default premium in relatively parsimonious manner.

25) To generate the synchronization in the financial market of U.S. and Korea, we linked the gap between the U.S. and Korea' s bond rates of the same maturity.

Figure 13. Corporate Bond Spread and Business Cycle



Note: Each line depicts corporate bond (AA-) spread from 3-year treasury bond (Blue, Left-axis) and HP-filtered GDP cycle (Golden, Right-axis). Shaded areas marks official recession periods by Statistics Korea

The bank loan rates are the interest rates that the private sector's agents (households and firms) actually face. Both banking rates share similar structure consist of the call rate and ten-year bond rate with different weights as household rate is closer to the longer bond rates and vice versa.

$$i_{HH,t} = \nu_{HH} i_t + (1-\nu_{HH}) i_{TB10Y,t} + \eta_{HH,t} \tag{40}$$

$$i_{Firm,t} = \nu_{Firm} i_t + (1-\nu_{Firm}) i_{TB10Y,t} + \eta_{Firm,t} \tag{41}$$

where $\eta_{HH,t}$ and $\eta_{Firm,t}$ are the spreads of household and firm respectively. These credit premiums are functions of the corporate bond spread in turn.

$$\eta_{HH,t} = (1-\rho_{HH})\bar{\eta}_{HH} + \rho_{HH}\eta_{HH,t-1} + \alpha_{HH}(\eta_{CB,t} - \bar{\eta}_{CB}) + u_{HH,t} \tag{42}$$

$$\eta_{Firm,t} = (1-\rho_{Firm})\bar{\eta}_{Firm} + \rho_{Firm}\eta_{Firm,t-1} + \alpha_{Firm}(\eta_{CB,t} - \bar{\eta}_{CB}) + u_{Firm,t} \tag{43}$$

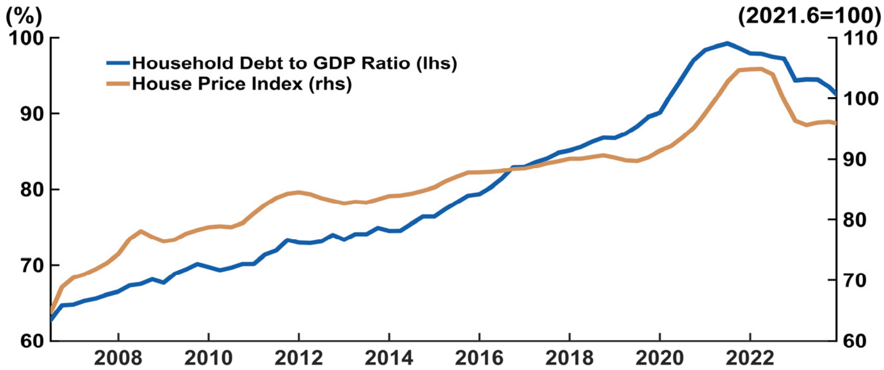
HOUSEHOLD DEBTS

The accumulation of household debt has been the major consideration from the financial stability perspective, while it is the public debt that most advanced economies have confronted.[26][27] Figure 14 highlights the persistent spiral of

26) Government debt-to-GDP ratio of Korea rose rapidly during the Covid-19 period, but it has never exceeded 50% much lower than the usual cases in many advanced economy.

household debt and housing price. In particular, Korea observed a drift of housing prices accompanied by that of debt-to-GDP ratio around 2019. Although the very recent trend has been declining, it's evident that the already high level of the ratio has become one of main concerns for the monetary policy pivot. Consequently it stresses the importance of taking the debt dynamics into the model.

Figure 14. Household Debt/GDP Ratio and Housing Price Index



Note: Each line depicts household Debt to GDP (Blue, Left-axis) and Housing Price (Golden, Right-axis)
Data source: ECOS

In the model, we designed the household debt-to-GDP ratio changes as a function of the output gap, housing price inflation, and the banking rate for households.[28)]

$$\frac{\Delta D_t}{4\,Y_t^{nominal}} = c_{debt} + \alpha_{debt,1}\hat{y}_t + \alpha_{debt,2}\Delta_4 \ln HPI_t + \alpha_{debt,3} i_{HH,t} + u_{debt,t} \tag{44}$$

We aimed the model to capture the lower frequency over the short-term dynamics of the debt-to-GDP ratio, by aligning the cyclical movement of the right-hand-side variables, which explains the reason for setting the housing price inflation as a year-over-year rather than quarter-over-quarter.

---

27) France shares a similar concern regarding financial stability from private sector and Banque de France actually came up with a satellite model of house prices, mortgage debt for the FR-BdF model. See Bove et al. (2020)

28) In the household debt-to-GDP ratio, we use quarterly nominal GDP level as its denominator.

Ⅳ. Results

1. Forecast Evaluation

To evaluate the economic forecasting performance of BOK-LOOK under the changed domestic and international economic conditions following COVID-19, we conducted a back-test for each quarterly forecast since 2021.

We used data vintages at each forecast point from 2021Q1 to 2024Q2, along with the assumptions and judgments from other departments of the bank.

Table 1 reports the forecast performance of BOK-LOOK on key macro variables compared to a benchmark model (BVAR).[29] Using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) to evaluate the forecast against the benchmark model, BOK-LOOK showed advancing performance in forecasting key macroeconomic variables such as growth and inflation.

Not only did the model provide strong midterm GDP predictions, but it also had significantly smaller forecast errors in inflation during the period of heightened uncertainty following COVID-19, indicating a high level of forecasting performance even under the changed economic environment after the pandemic.

Recall that the primary goal of this project was to develop a model capable of producing robust projections for key macroeconomic variables, including output, inflation, and the policy rate, even amid shifts in external and internal conditions. Stability in these forecasts is essential, as it provides policymakers with a clear baseline for evaluating economic trends and monetary policy decisions. Excessively volatile or overly sensitive projections, by contrast, risk amplifying uncertainty and undermining confidence in the decision-making process.

In particular, the model was designed to deliver a reliable baseline projection for the policy rate, which serves as an important signal to the public about the Bank of Korea's outlook on the economy. Achieving this requires stable forecasts for output and inflation, the two fundamental inputs systematically considered in

29) The benchmark Bayesian VAR(2) contains 11 key macro variables including GDP, inflation, interest rate and exchange rate. Three of them are U.S. observables.

standard monetary policy frameworks. As shown in Figure 15 to Figure 17, the projections for these variables tend to converge toward reasonable and interpretable midterm values, reflecting the model's capacity to filter through economic noise and focus on actionable insights. Consequently, the resulting policy rate paths effectively capture how the Bank of Korea interprets the surrounding economic environment, offering a coherent narrative for stakeholders and the public alike.

Table 1. Forecast Performance
(Ratio of RMSE relative to VAR) Since Covid-19

	GDP		CPI		Core	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
t+2	0.81	0.89	0.51	0.47	0.64	0.60
t+3	0.81	0.86	0.62	0.55	0.62	0.59
t+4	0.62	0.56	0.76	0.76	0.61	0.59
t+5	0.61	0.56	0.75	0.75	0.61	0.58
t+6	0.52	0.43	0.73	0.71	0.61	0.56
t+7	0.52	0.41	0.70	0.69	0.61	0.57

Note: RMSE/MAE normalized by that of BVAR(2) being 1. Therefore forecast performance of BOK-LOOK is superior over that of BVAR(2) if the number is lower than 1. Each row demonstrates the relative RMSE/MAE values at forecast horizons from two periods to 7 periods ahead.

Figure 15. Conditional Quasi-Forecasts on GDP Growth(Quarter-on-Quarter)

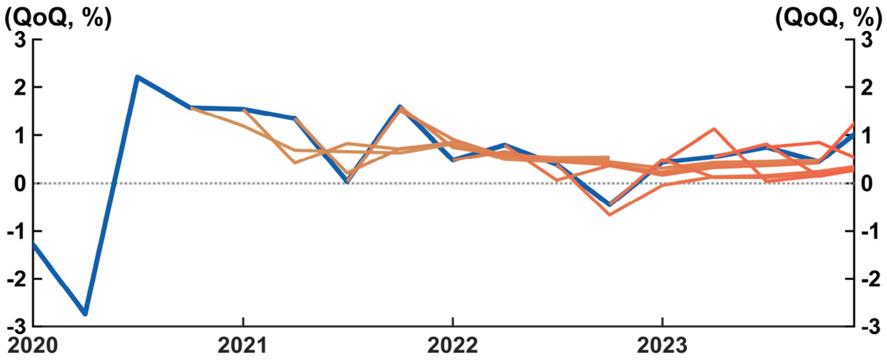


Figure 16. Conditional Quasi-Forecasts on Core CPI Inflation(Year-on-Year)

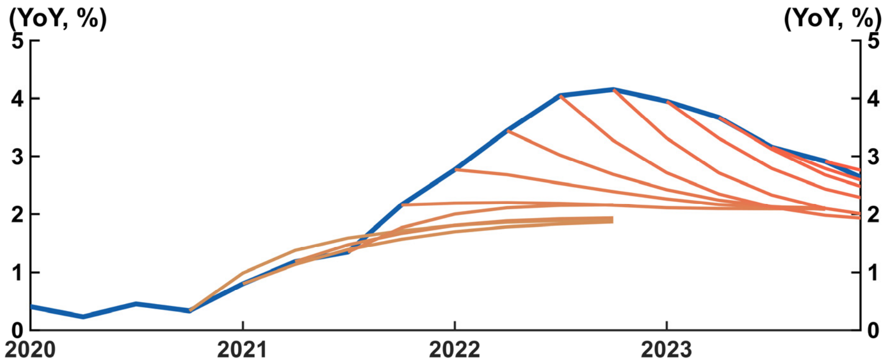
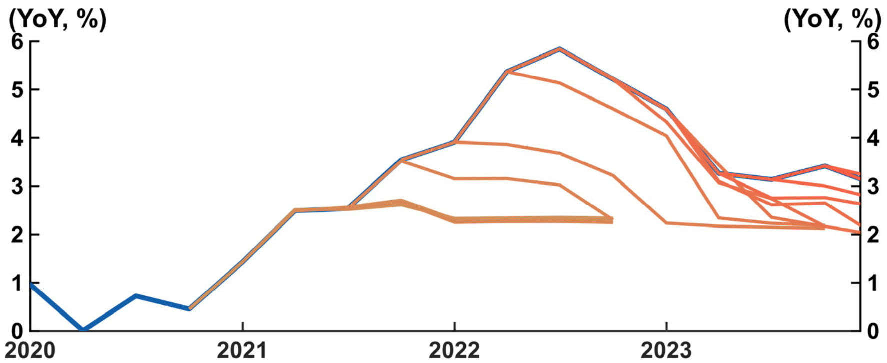


Figure 17. Conditional Quasi-Forecasts on CPI Inflation(Year-on-Year)



Note: Conditional forecast based on BOK's outlook premises at each vintage point (21Q1 to 23Q4). Blue lines depict reailzied observables, and from Golden to Red lines are Quasi-forecasts at each forecast point.

2. Impulse Response Analysis

This section examines the responses of domestic financial and economic variables to major domestic and international economic shocks using the BOK-LOOK model. The economic shocks considered include changes in policy interest rates in Korea and the United States, as well as an increase in international oil prices.

Korea's Monetary Policy Shock

When the base interest rate increases, several channels through which this affects major macroeconomic variables are activated. The initial impact is transmitted via market interest rates, which rise in response to the base rate hike. As market interest rates increase, borrowing costs for both consumers and businesses also rise, resulting in reduced consumption and investment activities. Consequently, the GDP gap moves in a negative direction, and the growth rate slows.

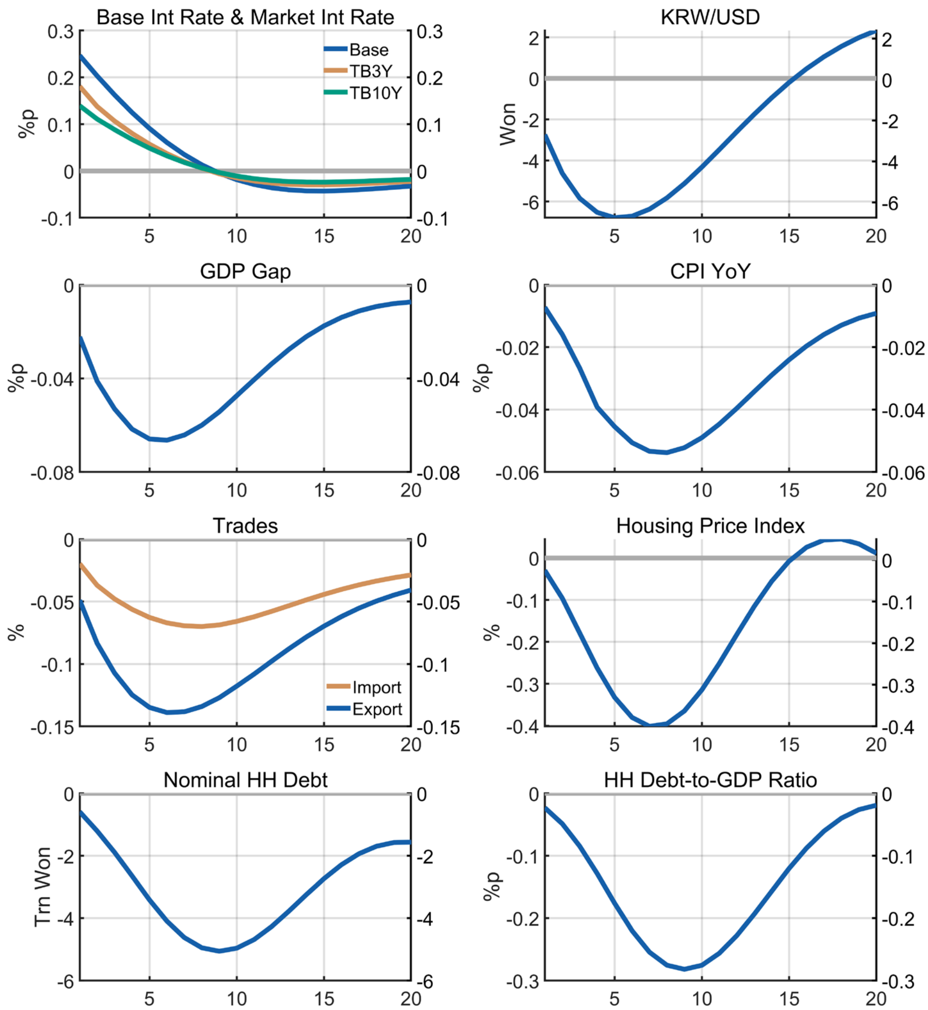
Specifically, a 25 basis point increase in the base rate raises market and lending rates, including household and corporate loan rates. This elevates the marginal cost of consumption and investment, leading to a contraction in both private consumption and business investments. As a result, the GDP gap decreases by up to 0.07 percentage points, while inflation, measured by the year-on-year change in the Consumer Price Index (CPI), declines by up to 0.05 percentage points after a delay.

The exchange rate channel also plays a significant role. As the base rate rises, the KRW/USD exchange rate tends to appreciate, following the Uncovered Interest Parity (UIP) condition. This appreciation makes Korean exports more expensive in the global market, which, combined with the reduction in aggregate domestic demand, leads to a decrease in total export demand. Furthermore, the stronger currency reduces import prices, contributing to a decline in overall inflationary pressures.

From a financial stability perspective, the effects of a base rate increase are notable in the housing market and household debt levels. Higher market interest rates directly impact mortgage rates, leading to a slowdown in housing price growth. Housing prices are estimated to decline by up to 0.4 percent, reducing the value of collateral held by households and limiting their borrowing capacity. This results in a decrease in household debt by approximately KRW 5.1 trillion, ultimately leading to a reduction in the household debt-to-GDP ratio by up to 0.3 percentage points.

Overall, the effects of monetary policy tightening can be summarized as follows: the initial increase in the policy rate leads to higher market interest rates, which affect both consumer and corporate behavior through increased borrowing costs. This, combined with changes in asset prices (such as housing) and exchange rate adjustments, influences aggregate demand and ultimately GDP and inflation.

Figure 18. Impulse Response Functions to Monetary Policy Shock (25bp)



U.S. Monetary Policy Shock

The Federal Reserve's policy rate hike affects the Korean economy through both financial synchronization and trade channels. When the U.S. policy rate increases, it directly influences Korean market rates due to strong financial linkages between the two countries, as well as indirectly through global trade impacts.

The financial synchronization channel is an important mechanism by which changes in the U.S. policy rate translate into changes in domestic financial conditions in Korea. A 25 basis point increase in the U.S. policy rate leads to an increase in U.S. market rates, which subsequently affects the Korean market through increased pressure on long-term interest rates, such as the 10-year government bond rate. The financial synchronization results in Korean long-term rates rising by up to 0.06 percentage points. As these rates rise, the cost of borrowing for both corporations and households in Korea increases, which suppresses investment and consumption.

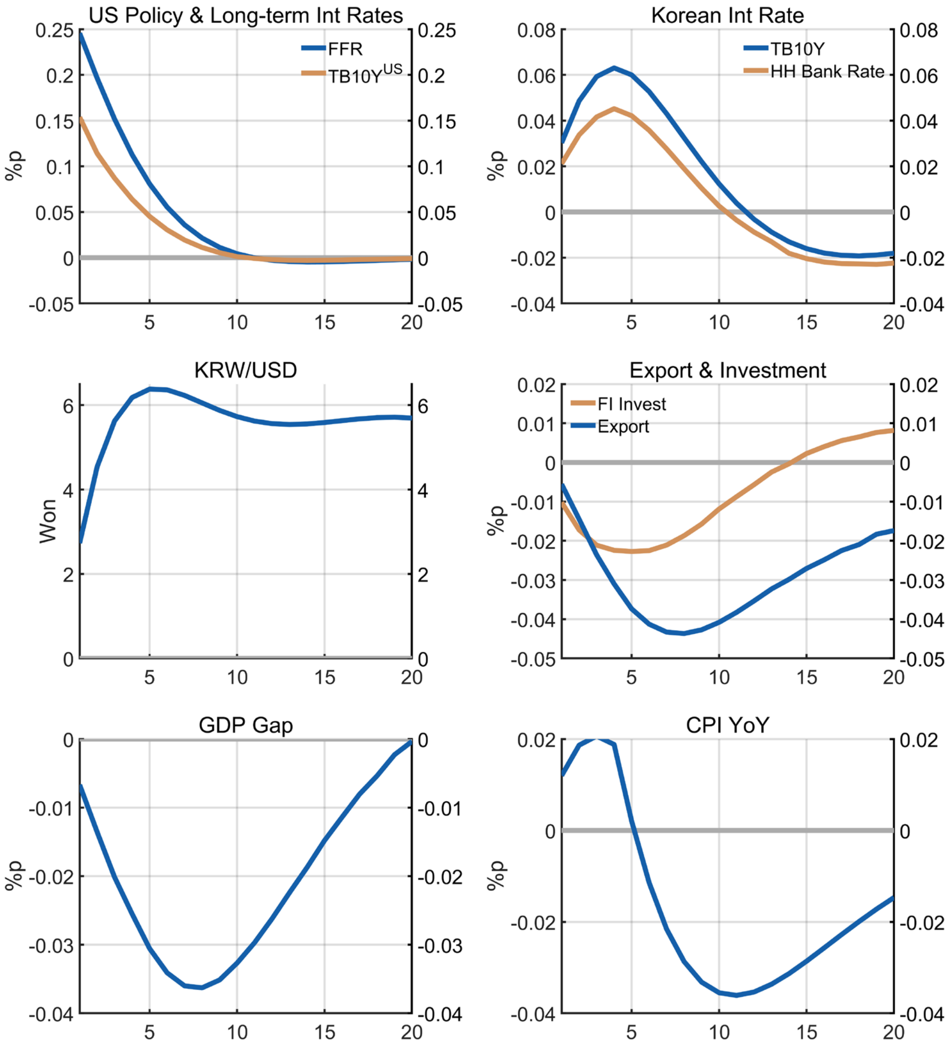
The exchange rate channel is also significant. With a higher U.S. policy rate, the KRW/USD exchange rate tends to depreciate as capital flows shift toward U.S. assets, attracted by relatively higher expected returns. The depreciation of the Korean won makes imports more expensive, initially pushing up domestic prices (CPI). However, the depreciation also has mixed effects on export dynamics. While a weaker currency could make Korean exports more competitive, the concurrent economic slowdown in the U.S. reduces overall global trade demand, ultimately reducing Korean export volumes.

In terms of the trade channel, the rise in the U.S. policy rate tends to slow down U.S. economic activity, reducing its demand for imports. This reduction negatively impacts Korean exports, particularly those destined for the U.S. market. As a result, the contraction in exports contributes to slower GDP growth in Korea, which decreases by up to 0.04 percentage points. The reduction in export demand also dampens facility investment by up to 0.02 percent, as companies face lower expectations for future sales.

The inflation response is twofold: initially, the depreciation of the Korean won pushes up inflation through higher import prices, particularly for energy and raw materials. However, as economic activity slows down due to reduced consumption, investment, and exports, inflationary pressures eventually decrease in the medium term. The net effect is an initial increase in the CPI followed by a subsequent moderation as the economy cools.

The overall impact of a U.S. policy rate hike is characterized by higher domestic interest rates, a depreciating currency, reduced export volumes, and an initial uptick in inflation followed by a slowdown. These factors collectively lead to a tightening of financial conditions, ultimately curbing GDP growth.

Figure 19. Impulse Response Functions to U.S. MP Shock (25bp)



International Oil Price Shock

An increase in international oil prices affects the domestic economy through multiple channels, including import prices, production costs, inflation, and overall demand. The shock impacts macroeconomic variables such as inflation, consumption, investment, and ultimately the GDP gap. In the BOK-LOOK model, these transmission channels are captured through interactions among various endogenous variables such as the Consumer Price Index (CPI), import prices, investment deflators, and exchange rates.

When international oil prices rise by 10%, the first effect is seen in import prices. The rise in oil prices increases import prices, directly affecting the CPI, causing an upward push in inflation by up to 0.16 percentage points in the short term.

In response, the central bank might increase the policy rate, which subsequently raises market interest rates, leading to a decline of up to 0.08 percent in consumption. Investment is also adversely affected. The increase in oil prices raises production costs, affecting construction investment and mechanical investment. Higher input costs reduce the profitability of investments, leading to lower capital accumulation. The investment deflator for construction and mechanical investment captures these dynamics by reflecting changes in the cost of investment inputs.

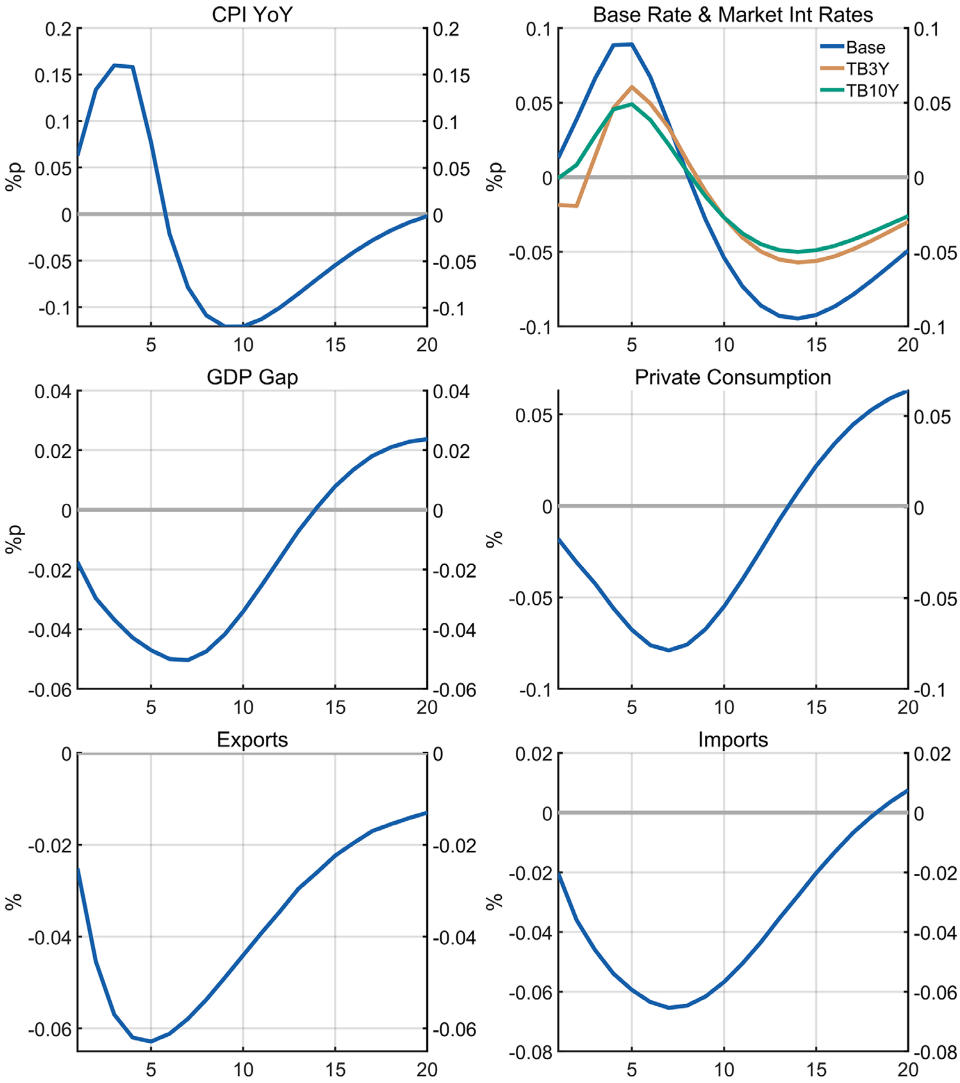
The combination of reduced consumption and investment leads to a contraction in overall demand, resulting in a decline in GDP growth. The model captures the GDP gap as a function of aggregate output and potential output. Higher oil prices, combined with increased interest rates and inflation, lead to a decrease in overall GDP by up to 0.05 percentage points. Imports also decline by up to 0.07 percent due to weaker domestic demand and higher costs.

Rising oil prices also have significant implications for global trade. Higher energy costs tend to dampen economic activity in major oil-importing countries such as China and the EU. As a result, global trade activity contracts, which further weakens Korean exports through reduced demand in these major trading partner countries. However, regions that are major oil producers (e.g., Rest of the World) benefit from rising oil prices, leading to an increase in their import gaps. This creates a mixed effect, where certain regions experience increased economic activity due to higher oil revenues. Nevertheless, the overall effect is a net global economic slowdown, resulting in reduced trade volumes, which negatively affects Korea's export performance.

In the medium to long term, inflationary pressures start to moderate as the higher interest rates implemented by the central bank begin to curb aggregate demand. As domestic growth slows, inflation declines due to reduced demand-pull pressures. The model's Phillips Curve helps explain the interaction between inflation and the output gap, reflecting how reduced economic activity eventually leads to lower inflation. The impact on exports is primarily negative, as rising energy costs dampen economic activity globally, leading to weaker demand from major trading partners.

To summarize, an increase in international oil prices leads to higher import prices, which directly contribute to an increase in the Consumer Price Index (CPI). This prompts the central bank to raise interest rates, leading to a reduction in private consumption and investment. The resulting contraction in global trade activity, particularly affecting major trading partners, further weakens Korean exports. Consequently, the GDP gap widens due to lower aggregate demand, and inflation eventually moderates in the medium term as economic activity slows.

Figure 20. Impulse Response Functions to Oil Price Shock (10%)



Ⅳ. Concluding Remarks

The BOK-LOOK model represents an important addition to the suite of macroeconomic tools used by the Bank of Korea. As a large-scale, semi-structural model, it combines theoretical consistency with empirical flexibility, offering a framework suitable for economic forecasting, scenario analysis, and policy evaluation. Drawing from the design principles of the FRB/US model, it has been adapted to reflect the distinct features of the Korean economy.

The model accounts for the interactions between domestic and global economic variables. With a structured representation of external trade and financial linkages, BOK-LOOK effectively captures Korea's sensitivity to external shocks. At the same time, its domestic blocks address critical issues such as household debt, housing prices, and their influence on monetary policy transmission, ensuring a comprehensive view of the economy.

Another key feature of the model is its flexibility and extensibility. It supports a wide range of applications, including medium-term projections, and counterfactual policy simulations.

Building on these features, the model has demonstrated dependable performance in projecting key macroeconomic indicators such as GDP growth and inflation. By generating endogenous policy rate projections that align with internal economic dynamics, it supports the Bank's quarterly economic outlook and provides valuable input to the Monetary Policy Committee's decision making.

Future improvements to BOK-LOOK will focus on addressing emerging challenges in the economic landscape. The modular architecture of the model will facilitate the addition of complementary features in the future, such as satellite models to enhance medium-term forecasting or optimal control-based rules for evaluating alternative monetary policies. The Bank plans to incorporate feedback from internal and external stakeholders to refine the model's functionality and scope.

In summary, BOK-LOOK enhances the Bank of Korea's analytical toolkit, serving as a core model for both forecasting and policy analysis. It provides reliable projections for understanding Korea's economic dynamics, particularly in an environment shaped by external dependencies and domestic financial complexities.

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Appendix. A Satellite VAR Models For Expectation Formation

VAR Satellite Model Specification

Let $X_t = \left[\pi_t\, \hat{y}_t\, R_t\right]'$ be a 3×1 vector of core variables-inflation, output gap and interest rate. Let $X_t^{ep} = \left[\pi_t^{ep}\, \hat{y}_t^{ep}\, R_t^{ep}\right]'$ be a 3×1 vector of shifting endpoints for each core variable defined as a long-run expectation at the current period, i.e. $z_t^{ep} \equiv \lim_{j\to\infty} E_t z_{t+j}$. In practice we use expert survey data on the longer-run expectation or deduce their values from a term-structure model.

Then we can construct a core VAR satellite model in a vector error-correction form as,

$$\Delta X_t = \Lambda_0 [X_{t-1}^{ep} - X_{t-1}] + \Lambda_1 \Delta X_{t-1} + \cdots + \Lambda_{p-1} \Delta X_{t-p+1} + u_t \tag{A.1}$$

The above setup clearly reveals the role of an endpoint. First, it serves as a converging point of each VAR component helping the VAR model shape reasonable long-run VAR expectation.

After establishing the core VAR model, we constructed an augmented VAR for an arbitrary variable of interest w_t to represent how model agents form expectation as

$$\Delta \begin{bmatrix} w_t \\ X_t \end{bmatrix} = \begin{bmatrix} \phi_0 & \Phi_0 \\ 0 & \Lambda_0 \end{bmatrix} \begin{bmatrix} w_{t-1} - w_{t-1}^{ep} \\ X_{t-1} - X_{t-1}^{ep} \end{bmatrix} + \begin{bmatrix} \phi_1 & \Phi_1 \\ 0 & \Lambda_1 \end{bmatrix} \Delta \begin{bmatrix} w_{t-1} \\ X_{t-1} \end{bmatrix} + \cdots + \begin{bmatrix} \phi_{p-1} & \Phi_{p-1} \\ 0 & \Lambda_{p-1} \end{bmatrix} \Delta \begin{bmatrix} w_{t-p+1} \\ X_{t-p+1} \end{bmatrix} + v_t \tag{A.2}$$

by restricting augmented VARs with block exogeneity, we can keep the core VAR coefficients identical and independent to the arbitrary variable w_t.

To utilize the VAR model for expectation formation, it is convenient to transform the model from the error-correction to a level representation with the augmented vector defined as $Z_t = \left[w_t \, \pi_t \, \hat{y}_t \, R_t\right]'$,

$$\Delta Z_t = \Gamma_0[Z_{t-1}^{ep} - Z_{t-1}] + \Gamma_1 \Delta Z_{t-1} + \cdots + \Gamma_{p-1}\Delta Z_{t-p+1} + u_t \tag{A.3}$$

$$\begin{aligned} Z_t &= \Gamma_0 Z_{t-1}^{ep} + [I - \Gamma_0 + \Gamma_1] Z_{t-1} + [\Gamma_2 - \Gamma_1] Z_{t-2} + \cdots \\ &\quad + [\Gamma_{p-1} - \Gamma_{p-2}] Z_{t-p+1} - \Gamma_{p-1} Z_{t-p} + u_t \\ &= \Gamma_0 Z_{t-1}^{ep} + A_1 Z_{t-1} + A_2 Z_{t-2} + \cdots + A_{p-1} Z_{t-p+1} + A_p Z_{t-p} + u_t \end{aligned} \tag{A.4}$$

And then turn it into a companion form as

$$\begin{bmatrix} Z_t \\ Z_{t-1} \\ \vdots \\ Z_{t-p+1} \end{bmatrix} = \begin{bmatrix} \Gamma_0 Z_{t-1}^{ep} \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} A_1 & A_2 & \cdots & A_p \\ & I_{p-1} \otimes I_n & & 0_{n(p-1)\times n} \end{bmatrix} \begin{bmatrix} Z_{t-1} \\ Z_{t-2} \\ \vdots \\ Z_{t-p} \end{bmatrix} + U_t \tag{A.5}$$

$$\Rightarrow z_t = \tilde{\Gamma}_0 z_{t-1}^{ep} + H z_{t-1} + U_t \tag{A.6}$$

with n stands for the dimension of the VAR model. Then from the one-step-ahead to long-run forecasts follow as

$$\begin{aligned} E_t z_{t+1} &= \tilde{\Gamma}_0 z_t^{ep} + H z_t, \\ E_t z_{t+2} &= \tilde{\Gamma}_0 E_t z_{t+1}^{ep} + H E_t z_{t+1} = \tilde{\Gamma}_0 E_t z_{t+1}^{ep} + H \tilde{\Gamma}_0 z_t^{ep} + H^2 z_t, \\ E_t z_{t+\tau} &= \sum_{j=0}^{\tau-1} H^j \tilde{\Gamma}_0 z_{t+j}^{ep} + H^\tau z_t \end{aligned} \tag{A.7}$$

Assuming that the endpoint processes follow random walk, then we get the τ step-ahead forecast as

$$E_t z_{t+\tau} = [I - H^\tau][I - H]^{-1} \tilde{\Gamma}_0 z_t^{ep} + H^\tau z_t \tag{A.8}$$

and the long-run forecast as

$$\lim_{\tau \to \infty} E_t z_{t+\tau} = [I - H]^{-1} \tilde{\Gamma}_0 z_t^{ep} \tag{A.9}$$

Note that the long-run forecast is solely dependent to the endpoints. It indicates that setting the endpoint can be critical given that the PAC has a term consisting of an infinite discounted sum of target variable changes.

PAC Equation Estimation Process with VAR Satellite Model

Once the VAR coefficients are estimated, estimation for a PAC equation can be described with the following steps,

1. Compute target variable $\{y_t^*\}_{t=1}^{T}$ from its behavioral equation
2. Estimate the augmented VAR model in companion form,

$$z_t = Hz_{t-1} \text{ where } z_t = \left[y_t^* \, \pi_t \, \hat{y}_t \, R_t\right]' \tag{A.10}$$

3. Estimate PAC equation

$$\Delta y_t = a_0(y_{t-1}^* - y_{t-1}) + \sum_{k=1}^{m-1} a_k \Delta y_{t-k} + E_{t-1}\sum_{k=0}^{INF} d_k \Delta y_{t+k}^* + \epsilon_t \tag{A.11}$$

where the long-run expectation term is a nonlinear function of $\{a_j\}_{j=0}^{m-1}$ and time discount rate β and the companion VAR coefficient H,

$$E_{t-1}\sum_{k=0}^{\infty} d_k \Delta y_{t+k}^* = A(1)A(\beta)[(\iota'_m(I_m - G)^{-1}) \otimes H'][I_{nm} - (G \otimes H')]^{-1}[\iota_m \otimes \iota_1]z_{t-1} \tag{A.12}$$

given that

$$G = \begin{bmatrix} 0 & I_{m-1} \\ -\alpha_m\beta^m - \alpha_{m-1}\beta^{m-1} & \cdots - \alpha_1\beta \end{bmatrix}, \tag{A.13}$$

$$A(L) = 1 + \alpha_1 L + \alpha_2 L^2 + \cdots + \alpha_m L^m \tag{A.14}$$

$$a_0 = A(1), \tag{A.15}$$

$$a_k = \sum_{j=k+1}^{m} \alpha_j \tag{A.16}$$

Appendix. B Toolkit for Conditional Forecasts

Models are primarily used to forecast endogenous variables in a way that ensures internal consistency among forecast results. For instance, if a model includes a structural Phillips curve, its projections for the output gap and inflation should be jointly determined. However, there are occasions when projections need to be made conditional on certain endogenous variables being fixed. This requirement often arises from the need to incorporate external information or to perform counterfactual analyses. For example, high-frequency data from the trade sector can provide valuable predictive insights for export and import forecasts, which might not be fully captured by the model.

To this end, we follow the spirit of Burgess et al. (2013) to invert the model and solve for the unobserved residuals consistent with observed data produced by the model simulation.

To specify predetermined paths for certain endogenous variables, the model's shocks (residuals) must be adjusted to identify a combination of endogenous variables that satisfies the given constraints. For instance, when the inflation forecast is fixed to align with projections from a nowcasting model, the forecast path for inflation, along with those for other endogenous variables in the model, is adjusted to account for their interactions with inflation. This approach ensures that the projections remain internally consistent while incorporating additional information or judgment.

Suppose we express our model as follows

$$x_t = f(x_{t-1}, z_t, z_{t-1}, u_t) \tag{B.1}$$

where x_t, z_t, u_t denote the vector of endogenous variables, exogenous variables and residuals respectively. Suppose further that some elements of x_{t+1} is predetermined,

$$x_{t+1} = [{x_{1,t+1}}' \, {x_{2,t+1}}']' \tag{B.2}$$

where we denote undetermined(free) endogenous variables as $x_{1,t+1}$ and the conditional ones as $x_{2,t+1}$.

The idea starts with the fact that technically, the endogenous variable $x_{2,t+1}$ is pinned down from the exogenous variables(predetermined from the model perspective) and residuals as

$$[x_{1,t+1}{}' \; x_{2,t+1}{}']' = f(u_{t+1}|x_t, z_{t+1}, z_t) \tag{B.3}$$

The given structure indicates that combinations of the residuals u_{t+1} can exist that spell out the certain value of $x_{2,t+1}$ conditioning on the values outside of the model(x_t, z_{t+1}, z_t). In other words, we can modify such mapping as

$$x_{2,t+1} = f(u_{t+1}; x_{1,t+1}|x_t, z_{t+1}, z_t) \tag{B.4}$$

which underscores that there can be some u_{t+1} inducing $x_{2,t+1}$.

The above expression directly becomes the target for conditioning, minimizing the distance between the left-hand and right-hand of the equation. However, note that there can be more than one set of residuals that satisfy the equation. Then how can we choose among the many residuals that give the same value of $x_{2,t+1}$? Ideally, it would be a vector of residuals in lines with the context of the forecast that a modeler posits.[30)]

Consequently, we numerically search for the u_{t+1} which solves a minimizing problem augmenting a term evaluating whether a u_{t+1} is likely to be realized based on the past experience,

$$\min_{u_{t+1}} \left(x_{2,t+1} - f(u_{t+1}; x_{1,t+1}|x_t, z_{t+1}, z_t)\right)^2 \times \left(\frac{\phi(\hat{u}; \hat{\Theta})}{\phi(u_{t+1}; \hat{\Theta})}\right)^{\alpha} \tag{B.5}$$

where $\phi(\,\cdot\,;\Theta)$ refers to a probability density function with its set of parameters Θ necessary for its identification and α is a hyperparameter between zero and one that governs the level of penalty in terms of likelihood measurement.[31)]

30) Instead, providing all possible u_{t+1} can be informative in terms of risk managements.

31) Currently we are approximating the p.d.f of the residuals with multivariate normal distribution even though residuals from a nonlinear model hardly follows it while bootstraping from their empirical p.d.f would arguably be a better alternative. Main reason is that there are some residuals with very large variance and/or trends yet filtered out which affects the forecast result critically. Therefore, we reconciled it with normal p.d.f approximation and let the selection of the residuals to be in neighborhood of estimated residuals. In addition, we manually selected the shocks(residuals) to be adjusted for the numerical solution and leaved most of the shocks remained unchanged based on the endogenous variables we target. For instance, we selected the import shock only if we conditioned import value.

Appendix. C Unobserved Component Model For Residuals

Due to their large scale and semi-structural specification, equations of FRB/US type models often leave additional information to their residuals. Modelers would select regressors go into an equation based on its theoretical\empirical relevance in a non-exhaustive way so that the equation remains interpretable and moderately sensitive. It could result in its residuals to be confounded with pure white-noises and trends unexplained by parsimonious regressors. Therefore there can be some room for improvement in forecasting performance by extracting the remaining trends and adding them to forecast paths.

Angelini et al. (2019) proposed a simple state space representation for the unobserved component of their residuals and to allow some residuals being constant even without the former approach.[32)]

Referring to the previous work, we set up an unobserved component model for an arbitrary residual $u_{i,t}$ as

$$u_{i,t} = \overline{u}_{i,t} + \epsilon_{i,t} - \epsilon_{i,t-1},\ \epsilon_{i,t} \sim N(0, \sigma^2_{\epsilon_i}) \tag{C.1}$$

$$\overline{u}_{i,t} = \phi \overline{u}_{i,t-1} + v_{i,t},\ v_{i,t} \sim N(0, \sigma^2_{v_i}) \tag{C.2}$$

in order to pick up the trend $\overline{u}_{i,t}$ and apply it to forecast paths.

If one produces a projection path without the fitted value of residual's trend, then it might end up with an unexplainable jump or drop due to the non-zero residuals at the end of the sample periods. In contrast, adding the residual's trend expected path smooths the forecast path as it gradually diminishes its value from the forecast path. Indeed, this methodology provides an unidentified component in a forecast path, weakening the interpretability of forecast results. However, at the cost of the semi-structure, we found the gain is prominent in level variables. GDP accounting is essential in our model since it determines GDP indirectly with the accounting. The problem occurs in the mismatch between the GDP observable and its explicit expenditure components due to inventories and statistical discrepancy which cannot be simply treated as zero as clearly seen in Figure C.1.

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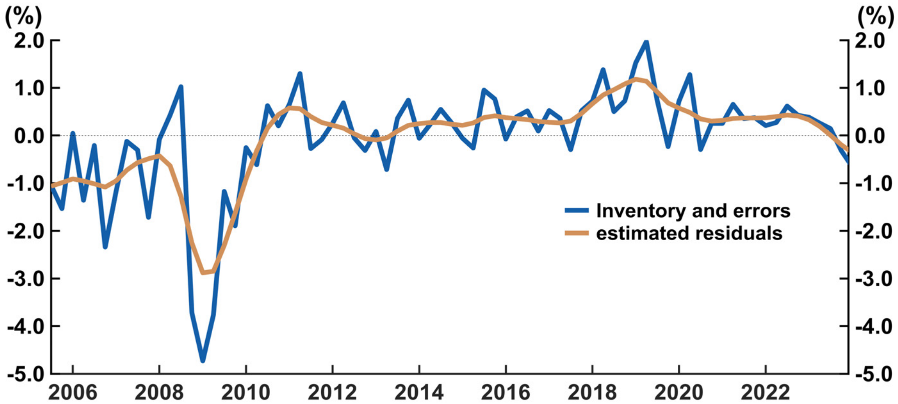
32) In Angelini et al. (2019), they keep some residuals like exchange rate and inventories. See the section 4.3.1 and Appendix C of their paper.

Figure C.1. Ratio of Inventory and Errors to GDP



Note: Blue line Gap between GDP and its expenditure components in real term, Golden line Estimated trends of the residuals
Data source: ECOS

Appendix. D Estimation Result Tables

Table 1. Estimates of the Foreign Block

Equation	Coefficients	Value
China	c^{CH}	0.0200
	τ_Y^{CH}	0.0500
	ρ_Y^{CH}	0.5943
	β_Y^{CH}	0.1673
	γ_Y^{CH}	0.0049
Japan	c^{JP}	0.0200
	τ_Y^{JP}	0.0500
	ρ_Y^{JP}	0.5819
	β_Y^{JP}	0.3054
	γ_Y^{JP}	0.0049
EU	c^{EU}	0.0200
	τ_Y^{EU}	0.0500
	ρ_Y^{EU}	0.6011
	β_Y^{EU}	0.3111
	γ_Y^{EU}	0.0049
EA	c^{EA}	0.0200
	τ_Y^{EA}	0.0500
	ρ_Y^{EA}	0.6151
	β_Y^{EA}	0.2706
	γ_Y^{EA}	0.0050

Table 2. Estimates of the Private Consumption Block

Equation	Coefficients	Value
Target Private Consumption	$\beta_{c,0}$	0.0038
	$\beta_{c,1}$	−0.1000
	$\beta_{c,2}$	−0.0384
	$\beta_{c,3}$	−0.1017
Growth Rate of Private Consumption	$\alpha_{C,0}$	0.0234
	$\alpha_{C,1}$	−0.1079
	$\gamma_{C,1}$	0.2005
	$\gamma_{C,2}$	−0.6000
	$\gamma_{C,3}$	0.0193
	$\gamma_{C,4}$	0.0200

Table 3. Estimates of the FI Investment Block

Equation	Coefficients	Value
Target FI Investment	$\beta_{I,0}$	−4.6240
	$\beta_{I,1}$	1.2121
	$\beta_{I,2}$	0.0274
Growth Rate of FI Investment	$\alpha_{I,0}$	0.1983
	$\alpha_{I,1}$	0.0248
	$\gamma_{I,1}$	0.6182
	$\gamma_{I,2}$	−0.1000
	$\gamma_{I,3}$	−0.0066

Table 4. Estimates of the Construction Investment Block

Equation	Coefficients	Value
Target Construction Investment	$\beta_{IH,0}$	4.8530
	$\beta_{IH,1}$	0.4187
	$\beta_{IH,2}$	0.0539
Growth Rate of Construction Investment	$\alpha_{IH,0}$	0.0267
	$\alpha_{IH,1}$	−0.2605
	$\gamma_{IH,1}$	0.2851
	$\gamma_{IH,2}$	−0.1000
	$\gamma_{IH,3}$	0.1671
	$\gamma_{IH,4}$	0.0629

Table 5. Estimates of the Government Consumption Block

Equation	Coefficients	Value
Target Government Consumption	$\beta_{G,0}$	−0.0974
	$\beta_{G,1}$	0.7913
	$\beta_{G,2}$	0.6582
Growth Rate of Government Consumption	c_G	0.0041
	$\alpha_{G,0}$	0.1697
	$\alpha_{G,1}$	−0.0440

Table 6. Estimates of the Export

Equation	Coefficients	Value
Target Export	$\beta_{X,0}$	4.1400
	$\beta_{X,1}$	1.6452
Growth Rate of Export	c_X	0.0085
	$\alpha_{X,0}$	0.0619
	$\alpha_{X,1}$	0.8192
	$\alpha_{X,2}$	0.0033
Export Demand	ζ^X_{US}	0.1800
	ζ^X_{EU}	0.0700
	ζ^X_{CH}	0.2000
	ζ^X_{EA}	0.1900
	ζ^X_{JP}	0.0500
	ζ^X_{RW}	0.3100

Table 7. Estimates of the Import

Equation	Coefficients	Value
Target Import	$\beta_{M,0}$	−4.5009
	$\beta_{M,1}$	1.3916
Growth Rate of Import	c_M	0.0028
	$\alpha_{M,0}$	0.1477
	$\alpha_{M,1}$	1.0108
	$\alpha_{M,2}$	−0.0062
Import Demand	ζ^M_C	0.2100
	ζ^M_I	0.2200
	ζ^M_{IH}	0.2200
	ζ^M_G	0.0800
	ζ^M_X	0.2900

Table 8. Estimates of the Core CPI

Equation	Coefficients	Value
Core CPI	ϕ_1	0.2500
	ϕ_2	0.1500
	ϕ_3	0.1000
	π^*	0.0200
	δ_1	0.7232
	δ_2	0.3164

Table 9. Estimates of the CPI

Equation	Coefficients	Value
Target CPI	ν_{cpi}	0.9790
Growth Rate of CPI	c_{cpi}	0.0011
	$\alpha_{cpi,0}$	0.0586
	$\alpha_{cpi,1}$	0.8040
	$\alpha_{cpi,2}$	0.0597

Table 10. Estimates of the Housing Price

Equation	Coefficients	Value
Target Housing Price	$\beta_{hpi,0}$	0.1693
	$\beta_{hpi,1}$	0.9539
	$\beta_{hpi,2}$	−4.5095
Growth Rate of Housing Price	c_{hpi}	0.0167
	$\alpha_{hpi,0}$	0.0405
	$\alpha_{hpi,1}$	0.5029
	$\alpha_{hpi,2}$	−1.4996

Table 11. Estimates of the Export Price

Equation	Coefficients	Value
Target Export Price	$\beta_{px,0}$	0.1890
	$\beta_{px,1}$	1.0000
Growth Rate of Export Price	c_{px}	0.0016
	$\alpha_{px,0}$	0.0982
	$\alpha_{px,1}$	0.3487
	$\alpha_{px,2}$	0.1381

Table 12. Estimates of the Import Price

Equation	Coefficients	Value
Target Import Price	$\beta_{pm,0}$	−4.6222
	$\beta_{pm,1}$	1.0651
	$\beta_{pm,2}$	0.4391
Growth Rate of Import Price	$c_{pm,0}$	0.0013
	$\alpha_{pm,0}$	0.1813
	$\alpha_{pm,1}$	0.8794
	$\alpha_{pm,2}$	0.2883

Table 13. Estimates of the Exchange Rate

Equation	Coefficients	Value
Exchange Rates	$\alpha_{EXR,0}$	0.8814
	$\alpha_{EXR,1}$	7.0142
	$\alpha_{EXR,2}$	4.0000
	c_{UIP}	0.0812

Table 14. Estimates of the Monetary Policy Rule

Equation	Coefficients	Value
Monetary Policy Rule	ϕ_i	0.85
	ϕ_π	1.5
	ϕ_{y_t}	0.5
	π^*	0.02

Table 15. Estimates of the Corporate Bond Rates

Equation	Coefficients	Value
Corporate Bond Rates	ρ_{CB}	0.5626
	$\bar{\eta}_{CB}$	0.0003
	α_{CB}	−0.0487

Table 16. Estimates of the Bank Loan Rates

Equation	Coefficients	Value
Bank Loan Rates	ν_{HH}	0.36
	ρ_{HH}	0.6066
	$\bar{\eta}_{HH}$	0.0038
	α_{HH}	0.2114
	$\bar{\eta}_{CB}$	0.0003
	ν_{Firm}	0.64
	ρ_{Firm}	0.7025
	$\bar{\eta}_{Firm}$	0.0013
	α_{Firm}	0.1607

Table 17. Estimates of the Debt to GDP Ratio

Equation	Coefficients	Value
Debt to GDP Ratio	c_{debt}	0.0208
	$\alpha_{debt,1}$	0.0118
	$\alpha_{debt,2}$	0.1204
	$\alpha_{debt,3}$	−0.8556

<Abstract in Korean>

BOK-LOOK: 우리나라 개방경제와 통화정책을 위한 준구조 모형

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본 논문은 한국은행의 경제전망체계 고도화를 지원하기 위하여 개발한 대규모 준구조 모형 BOK-LOOK을 소개한다. 본 모형은 미 연준의 FRB/US 모형에 기반하여 이론적 일관성과 실증적 유연성을 결합하고 한국 경제의 특수한 환경에 적합하도록 설계되었다.

BOK-LOOK은 글로벌 경제와 고도로 연계된 한국 경제에서 거시경제적 역학과 통화정책의 효과를 분석하기 위한 체계적 틀을 제공한다. 명시적 미래기대 형성과 거시경제와 금융 부문 연결성을 모형에 반영함으로써, 본 모형은 글로벌 경제 변화에 따른 한국 경제의 주요 거시경제 변수에 대한 정합적인 전망을 생성하며, 다양한 시나리오 분석을 지원한다.

베이지안 방법론을 통해 모수 추정의 선험적 정합성과 관측된 데이터와의 일관성을 도모하고 이에 따라 모형의 결과가 단기적 역학과 장기적 공적분 관계를 모두 포착하고 다양한 경제적 상호작용을 통합할 수 있도록 하였다. 그 결과, 모형이 주요 거시변수인 생산, 인플레이션, 이자율 등에 대해 내생적 전망을 제공하여 외부 충격과 통화정책의 영향을 평가하기 위한 시나리오 분석뿐만 아니라 기본 전망 경로를 안정적으로 생성하는 데 효과적임을 보였다.

이러한 특성들은 BOK-LOOK이 한국은행의 분석 도구로서 중요한 역할을 수행하며, 복잡한 경제 환경 속에서 데이터 기반의 정책 결정을 지원하는 데 기여할 것으로 기대된다.

핵심 주제어: FRB/US, 준구조 모형, 경제전망, 정책분석, 통화정책

JEL Classification: C3, E17, E37, E52, E58

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